

ESSAYS ON INDEX OWNERSHIP AND MARKET MICROSTRUCTURE

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By

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ABSTRACT

This dissertation studies how index ownership affects market quality. Using a quarterly panel of 799,337 observations covering 26,395 U.S. firms from 1993Q1 to 2023Q4, it examines liquidity, volatility, and stock price informativeness measured using return synchronicity.

The first chapter combines two-way fixed-effects regressions with instrumental-variable (IV) estimates exploiting Russell 1000/2000 reconstitution cutoffs and flexible nonlinear specifications. Baseline estimates indicate that greater index ownership is associated with lower volatility and lower price informativeness, while the liquidity effect is less uniform. The IV results show that these effects vary across firm environments: index-induced ownership changes improve liquidity and reduce price informativeness near the Russell 2000 cutoff, whereas the most robust effect near the Russell 1000 cutoff is a reduction in volatility. Nonlinear estimates further indicate that market-quality effects are not monotonic, with the most favorable outcomes occurring at intermediate ownership levels and the clearest curvature appearing in volatility.

The second chapter examines heterogeneity across firm size and business-cycle conditions. The informativeness effect is concentrated among smaller firms, while liquidity and volatility responses vary across size groups and macroeconomic regimes. Overall, the findings suggest that index ownership entails a context-dependent trade-off between liquidity provision, volatility, and information production rather than a uniform effect on market quality.

INDEX WORDS: Index Ownership, Passive Investing, Market Quality, Liquidity, Volatility, Price Informativeness, Market Microstructure, Stock Synchronicity

DEDICATION

To my family, whose love, patience, and unwavering support made this dissertation possible.

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CHAPTER 1

HOW MUCH INDEXING IS GOOD FOR THE MARKET?

1.1 INTRODUCTION

The rise of index investing has reshaped the composition of equity ownership and raised new questions about market quality. The growth of index investing is remarkable in the U.S. stock market. Introduction of the first index mutual funds was in the 1970s, followed by index ETFs in the 1990s. By the end of 2020, total net assets of these two index funds had grown to \$9.96 trillion,¹ which is the total net asset value of the 2010 fund market. In the sample studied in this chapter, average index ownership increase from low single digits in the late 1990s to roughly 11 – 12% by 2023, indicating a broad-based shift rather than a tail-driven phenomenon.

Traditional finance theory has emphasized that active investors shape price informativeness, volatility, and liquidity through information production, monitoring, and trading. Much of the market-quality literature therefore focuses on how informed trading and active monitoring shape price discovery and trading costs. The rise of passive ownership changes this benchmark by reallocating ownership toward index funds with limited incentives to gather firm-specific information. Understanding how index ownership affects market quality is thus a natural extension of the active-investor paradigm and central to modern asset markets.

¹Source: Investment Company Institute (2021), *2021 Investment Company Fact Book*, Table 42, pp. 48–49. The figures refer to U.S. equity index mutual funds and exchange-traded funds.

While passive vehicles expand access to diversified portfolios and reduce trading costs, they may also alter incentives for information production and liquidity provision. As passive ownership grows, a larger share of trading can be driven by index flows rather than firm-specific signals, potentially affecting price discovery and volatility. These competing forces make it unclear whether greater index ownership improves or impairs liquidity, volatility, and price informativeness. Taken together, these facts motivate a systematic investigation of how index ownership affects liquidity, volatility, and price informativeness.

1.1.1 RELATED LITERATURE

A large literature emphasizes the role of active investors in producing and incorporating information into asset prices. In models with costly information acquisition, individuals acquire information only when the expected benefits of informed trading compensate for its acquisition cost (Grossman and Stiglitz, 1980). Price formation models describe how private information is impounded through trading (Kyle, 1985; Glosten and Milgrom, 1985). Market microstructure theory links asymmetric information to bid–ask spreads and liquidity, highlighting the importance of trading intensity and informed participation (Amihud and Mendelson, 1986; O’Hara, 1995). A related literature shows that arbitrage corrects mispricing but is limited by capital and risk constraints (Shleifer and Vishny, 1997). This active-investor benchmark provides the natural reference point for evaluating the rise of passive ownership.

As passive investing has expanded, a growing literature has examined its implications for market quality. Index funds have become increasingly popular in part because their low fees help generate higher average net returns than those of actively managed mutual funds. On the positive side, passive vehicles can deepen liquidity and reduce transaction costs by supplying predictable demand and supporting trading volume

(Amihud and Mendelson, 1986; Admati and Pfleiderer, 1988). Empirical work documents that passive ownership can change trading activity and liquidity following index inclusion (Hjalmarsson, 2008) and that some index additions improve market depth (Harris and Gurel, 1986; Shleifer, 1986). On the negative side, ETFs may transmit shocks across securities and increase return comovement and volatility (Ben-David, Franzoni, and Moussawi, 2018), while higher passive shares can reduce incentives for information production and weaken price efficiency (Grossman and Stiglitz, 1980; Petajisto, 2013; Kozak, Nagel, and Wang, 2020). More broadly, Lamont (2026) argues that indexing does not mechanically reduce market efficiency because price discovery depends on the incentives, capital, and information of investors who remain active. These mixed findings motivate closer attention to the quasi-experimental evidence from index reconstitutions.

A large empirical literature examines index inclusion and demand-driven price effects. Early studies document price and volume changes around index additions, consistent with index fund demand shocks (Harris and Gurel, 1986; Shleifer, 1986). Subsequent research shows that arbitrage only partially offsets these shocks, implying persistent price impacts and downward-sloping demand curves for stocks (Wurgler and Zhuravskaya, 2002). Evidence from index revisions further shows that inclusion affects liquidity and trading behavior, reinforcing the role of passive flows in shaping market dynamics (Hjalmarsson, 2008). These findings motivate using index reconstitution as a quasi-experimental source of exogenous variation in passive ownership.

Taken together, the literature highlights both benefits and costs of passive ownership, but it provides limited guidance on the overall equilibrium effects on market quality. On the one hand, more index trading can lead to liquidity gain by reduced trading costs. On the other hand, a rising passive ownership may weaken incentives for information production and price discovery. These opposing channels imply that

the relationship between indexing and market quality may be non-monotonic, and beyond some level the decline in information production could dominate the liquidity gains. As a result, a central open question is whether there exists an interior level of indexing that balances these forces.

The core research question is: how much indexing is good for the market? Market quality is measured using three complementary proxies: Amihud illiquidity, realized volatility, and a synchronicity-based price informativeness measure. A related question is whether the effect is monotonic or varies with ownership intensity. Over the same horizon of increasing index ownership, market-quality measures exhibit pronounced cyclical dynamics, which complicates inference. Amihud illiquidity spikes in recessions while the median remains near zero, suggesting that stress episodes are driven by a tail of illiquid observations. Volatility rises sharply in downturns and falls in expansions, and price informativeness dips during recessions before rebounding in recoveries. These patterns imply that the effects of indexing may be state-dependent and non-monotonic.

1.1.2 EMPIRICAL STRATEGY AND CONTRIBUTION

This paper contributes to the literature on passive investing and market quality by providing new empirical evidence on how index ownership shapes the functioning of equity markets. The analysis draws on quarterly holdings data on U.S. equity index mutual funds and exchange-traded funds to measure index ownership, together with daily CRSP data on U.S. publicly traded firms to construct market-quality measures and firm-level controls. Aggregating the CRSP variables to the quarterly frequency produces a panel of 799,337 observations spanning 1993Q1–2023Q4. The empirical strategy follows a standard progression from descriptive evidence to causal identification and functional-form testing.

First, the study develops a direct measure of index ownership using fund-level holdings from index mutual funds and ETFs, rather than relying solely on aggregate passive-investment flows, index membership, or broad institutional ownership. Using this measure, baseline fixed-effects regressions document partial correlations between index ownership and each market-quality outcome. This provides a comprehensive analysis of how index ownership affects multiple dimensions of market quality—liquidity, volatility, and price informativeness—within a unified empirical framework. Prior work often isolates a single outcome or mechanism within a limited time horizon; in contrast, this study evaluates a broader set of market-quality measures and interprets them jointly. This design allows the analysis to assess whether improvements in one dimension are accompanied by deterioration in another, which is central to the equilibrium tradeoff implied by passive ownership.

Second, the study contributes new causal evidence by using the annual Russell 1000/2000 reconstitution cutoffs as instruments for index ownership. Prior studies have used this source of variation to identify the effects of passive or institutional ownership on corporate governance (Appel et al., 2016), payout policy (Crane et al., 2016), and price informativeness (Sammon, 2024). The identification strategy is therefore established; the contribution of the present study is to compare the causal effects of index-induced ownership shocks across the two cutoff environments, which generate different ownership changes. Because firms near each cutoff are similar in size but receive different index weights, the cutoff provides quasi-exogenous variation in index ownership. The results show that the causal effects are cutoff-specific rather than uniform: volatility effects are most pronounced near the Russell 1000 cutoff, whereas liquidity and price-informativeness effects are strongest near the Russell 2000 cutoff.

Third, the study adds a nonlinear dimension to the literature by examining whether the market-quality effects of index ownership change across the ownership distribution rather than imposing a constant marginal effect. Quadratic and spline specifications allow the ownership–market-quality relationship to change across the observed ownership distribution and assess whether the combined effects across liquidity, volatility, and price informativeness produce an interior range of ownership associated with stronger overall market quality. To account for endogeneity, the nonlinear analysis is extended to an instrumental-variable (IV) setting by instrumenting both ownership and its square to allow curvature in causal effects. This approach evaluates whether the nonlinear pattern remains after both functional-form flexibility and endogeneity are taken into account. The prevailing empirical literature largely focuses on average linear effects, even though theory predicts that passive investing can improve liquidity while weakening incentives for information production.

This chapter is organized as follows. Section 1.2 describes the data and shows that index ownership increased broadly from the late 1990s to 2023. Section 1.3 presents the baseline fixed-effects results, which associate greater index ownership with higher illiquidity, lower volatility, and lower values of the price-informativeness proxy. Section 1.4 presents the IV analysis and shows that the effects differ across cutoffs: ownership reduces volatility near the Russell 1000 cutoff and improves liquidity while reducing price informativeness near the Russell 2000 cutoff. Section 1.5 examines nonlinear effects using quadratic specifications and a composite market-quality score, identifying an interior ownership range associated with the most favorable market-quality outcomes. Section 1.6 evaluates these nonlinear patterns using quadratic and spline IV models, with the strongest causal evidence appearing in the volatility channel. Section 1.7 concludes.

1.2 DATA

1.2.1 DATA DESCRIPTIONS

The empirical analysis uses a quarterly panel of U.S. publicly traded stocks to examine the relationship between index-fund ownership and stock-level market outcomes. Index ownership is constructed from quarter-end portfolio holdings reported by U.S. equity index mutual funds and exchange-traded funds, while market-quality outcomes and firm-level controls are calculated from daily CRSP observations for the underlying stocks. The funds provide the ownership information but are not the units of analysis; the data are organized at the stock-quarter level. The resulting panel includes 799,337 observations spanning 1993Q1 through 2023Q4, with 26,395 distinct CUSIPs and 124 quarter-end dates. Each observation is keyed by stock identifier and quarter end and contains a primary transaction-cost measure alongside controls for size, trading activity, liquidity, and ownership.

The sample is an unbalanced panel. Firms enter and exit over time, and variable coverage varies across ownership measures and market-quality proxies. For descriptive statistics, the reported moments are computed using non-missing and non-infinite values within each column. This approach preserves the broad panel structure while ensuring that summary statistics are not distorted by computational artifacts.

Two features of the data merit emphasis at the outset. First, the long time coverage allows the analysis to span multiple business cycles, including recession and recovery periods, which is essential for later heterogeneity tests. Second, the distribution of key variables is highly skewed, which implies that summary statistics and regression estimates should be interpreted with attention to dispersion and tails rather than only to central tendencies.

Index ownership and institutional ownership are constructed from mutual fund holdings reported in Thomson Reuters Mutual Fund Holdings (S12), accessed through WRDS. These holdings data are observed at quarter-end frequency and provide direct measures of fund-level portfolio positions. They cover a broad set of domestic mutual funds and ETFs with long time-series availability. This coverage supports analysis of ownership changes over multiple business cycles.

Market-quality outcomes and firm controls are derived from the CRSP daily stock file, including bid and ask prices, trading volume, market capitalization, and return. The CRSP records are observed daily and then aggregated to the quarter level for alignment with ownership data. Daily variables are therefore summarized within quarter windows to construct stock-quarter measures. This aggregation reduces high-frequency noise while maintaining comparability across firms and time.

Daily CRSP data are converted into quarterly measures using within-quarter aggregation rules that match each construct. The Amihud illiquidity measure is computed from daily absolute returns scaled by daily dollar volume and then averaged over trading days in the quarter. Economically, this measure captures the price impact of trading: higher values indicate that a given amount of trading is associated with larger price changes, implying lower liquidity and higher trading costs. Volatility is computed as the standard deviation of daily returns within the quarter, yielding a realized volatility measure for each stock-quarter. It captures price variability rather than transaction costs.

The price informativeness proxy is based on stock return synchronicity, which measures the extent to which a stock's returns co-move with common market factors; lower synchronicity indicates greater firm-specific information in prices. In other words, lower common-factor comovement suggests that a larger share of price variation reflects firm-specific information. The measure is then summarized at the quar-

terly level, preserving the sign and scale implied by its transformation. These measures capture distinct dimensions of market quality and therefore need not move together: a stock may be liquid but volatile, or actively traded while its price contains relatively little firm-specific information. Examining them jointly allows the analysis to distinguish the effects of index ownership on trading conditions, price risk, and information incorporation.

With these transformations in place, the merged dataset combines a holdings-based ownership panel with a trading-based market-quality panel into a unified quarterly file. The alignment choice treats ownership as a quarter-end state variable while market quality is measured over the same quarter. This timing convention reduces ambiguity about causal ordering in descriptive analyses. It also matches the fixed-effects framework used in later regressions.

1.2.2 KEY TRENDS

Liquidity and volatility measures are well-behaved in central tendency but display long upper tails. Amihud illiquidity measure has a mean of 0.136 and a median of 0.0128, a ratio that indicates heavy right skew typical of liquidity metrics. Realized volatility averages 0.182 with a median of 0.150, but the maximum exceeds 39, which likely reflects rare, high-stress quarters or firm-specific events. These tails motivate the use of fixed effects and robust inference in later regressions. Price informativeness displays a nearly symmetric spread around its mean, with a minimum of -6.91 and a maximum of 6.91. The interquartile range spans 0.385 to 2.571, indicating that most observations cluster around moderate informativeness values, while the tails capture extreme comovement or idiosyncratic dynamics.

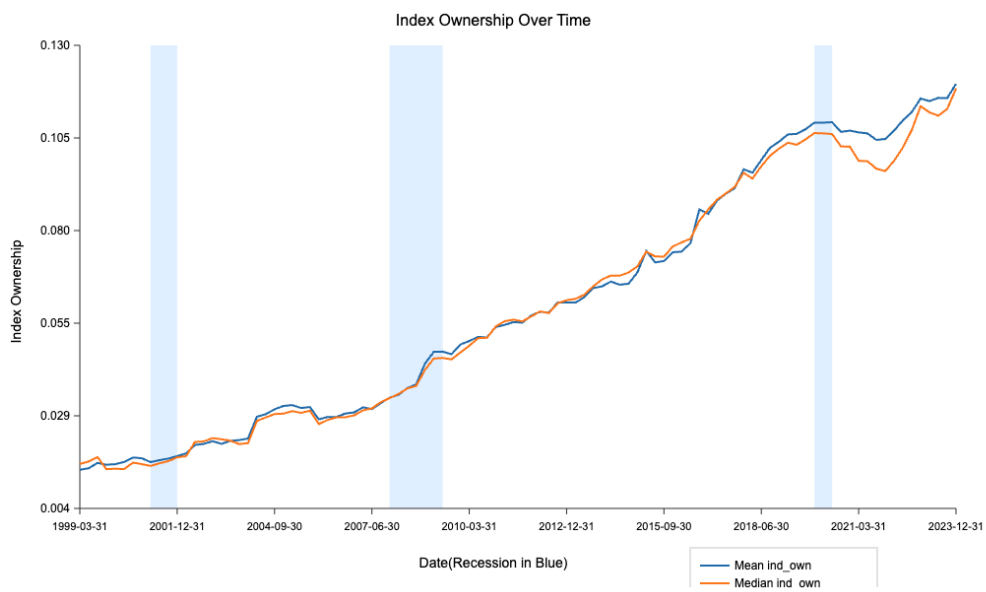
Ownership variables are more concentrated. Index ownership averages 0.069 with an interquartile range from 0.026 to 0.095. Institutional ownership has a mean of

0.469 and a median of 0.476, indicating broad participation of institutions across the sample. The ownership distributions underscore that even modest changes in index holdings can be economically meaningful when they occur in large firms or in periods of heightened market stress.

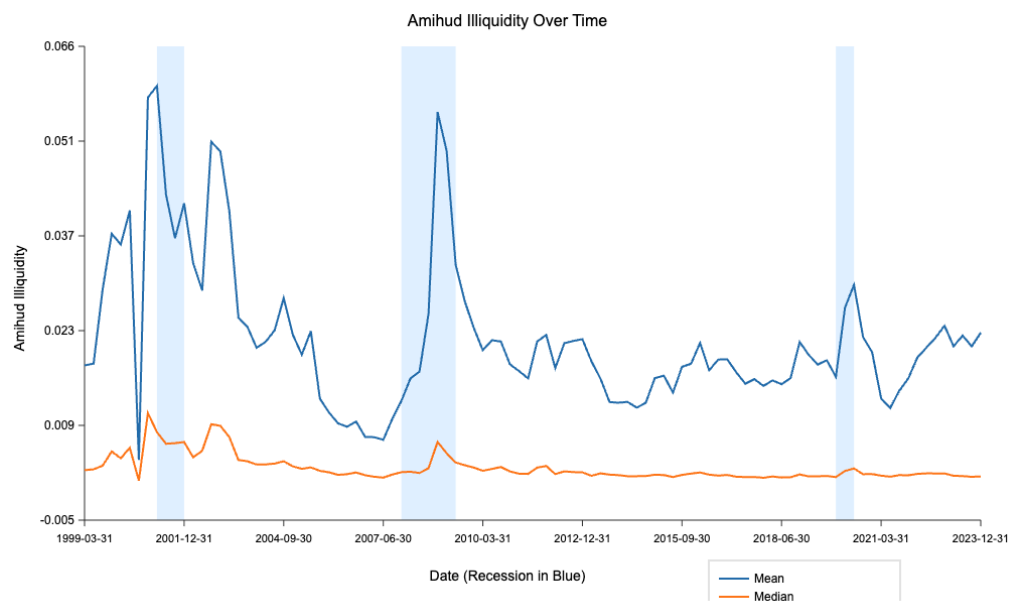
The constructed panel provides a long-horizon view of firm-level market quality, trading activity, and ownership structure. The data are rich in time coverage and cross-sectional breadth, yet they display notable skewness and ownership-related missingness. These features inform both the specification of baseline regressions and the interpretation of heterogeneity analyses in subsequent chapters.

Figure 1.1 show a steady increase in index ownership over the full sample, with mean and median ownership rising from low single digits in the late 1990s to roughly the low teens by 2023. Recession windows in 2001, 2007–2009, and 2020 are visible as brief slowdowns but do not reverse the long-run upward trend. The closeness of the mean and median lines indicates that the increase is broad-based rather than driven by a narrow tail of firms. This pattern suggests that index participation has deepened across the cross section even as macro conditions have fluctuated.

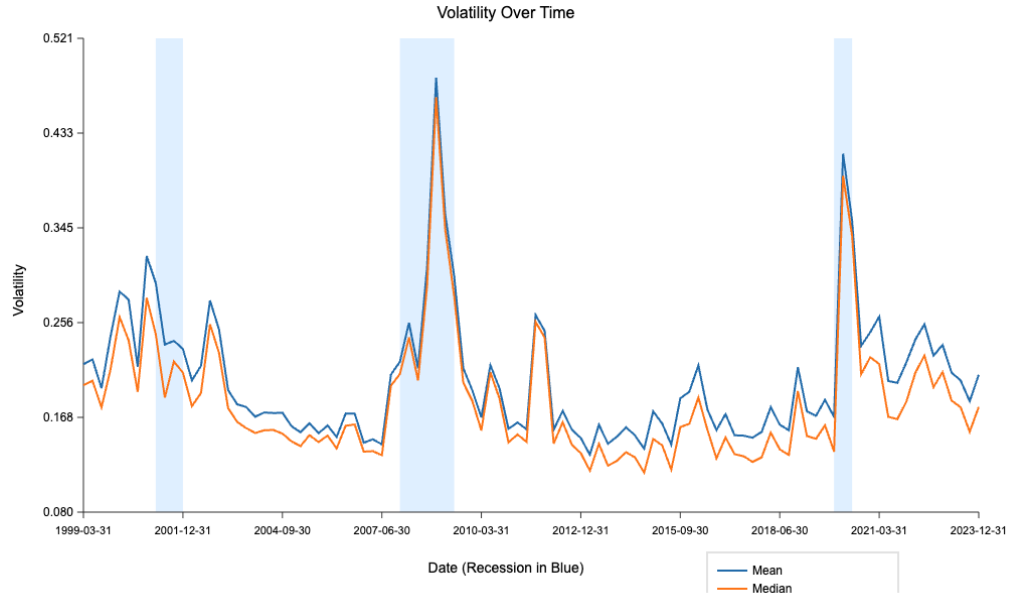
Liquidity, informativeness, and volatility move sharply with the cycle, but their behaviors differ in level shifts and distributional shape. Amihud illiquidity spikes in recessions while the median remains near zero, implying that stress episodes are driven by a tail of illiquid observations rather than a uniform shift in the cross section. The post-2009 decline in the mean suggests a gradual easing of liquidity frictions even as index ownership expands, which points to a coexistence of indexing growth and improved trading conditions outside stress windows. Price informativeness dips during recessions and then rebounds, and the close alignment of mean and median implies a broad distributional shift rather than a tail event. Volatility rises sharply in recessions and falls during expansions, with spikes that appear economy-wide rather



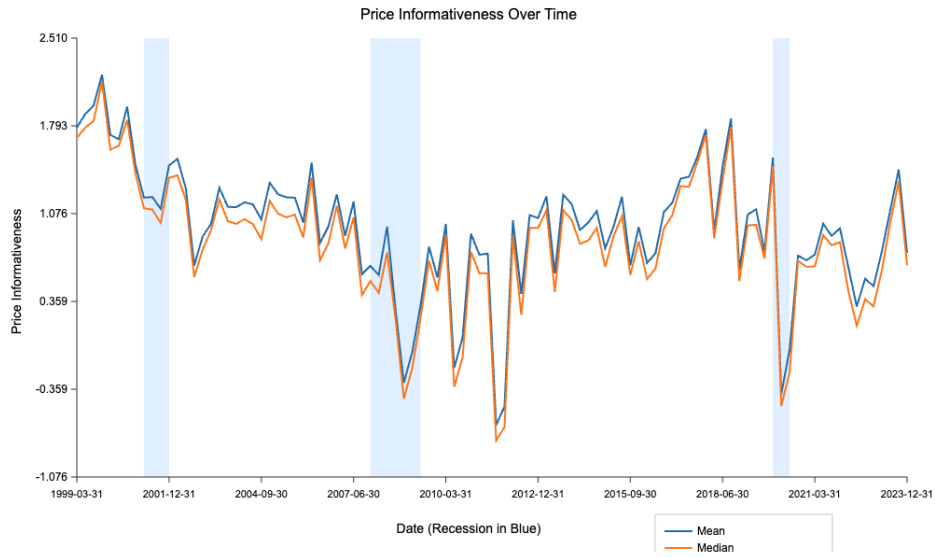
(a) Index Ownership



(b) Illiquidity



(c) Volatility



(d) Price Informativeness

Figure 1.1: Index Ownership and Market Quality over Time

Notes: Data Sources: WRDS-Thomson S12 and CRSP; Recessions are highlighted in blue.

than concentrated in a small subset of firms, indicating that macro shocks dominate firm-specific dispersion in those periods.

1.3 BASELINE ANALYSIS

1.3.1 PANEL REGRESSION MODEL

This section establishes the baseline panel specification that links index ownership to firm-level market-quality outcomes. The purpose is to provide a transparent descriptive benchmark that conditions on firm and time fixed effects within a long quarterly panel. The baseline regression lays the empirical foundation for the later identification and nonlinearity analysis. These estimates summarize partial correlations that are useful for comparison across outcomes and subsamples.

The baseline models are estimated separately for each outcome using a two-way fixed-effects design with two-way clustered standard errors. Firm fixed effects absorb time-invariant heterogeneity in trading frictions, while time fixed effects capture common macro and market-wide shocks. Standard errors are clustered by firm and date to accommodate serial correlation within firms and cross-sectional dependence across firms in the same quarter. This specification is held constant across outcomes to ensure comparability of the index-ownership coefficient. It therefore provides a disciplined platform for subsequent extensions and robustness analysis.

The baseline panel regression model is constructed as:

$$Y_{i,q} = \alpha_i + \lambda_q + \beta IndexOwn_{i,q} + \gamma X'_{i,q} + \epsilon_{i,q} \quad (1.1)$$

where $Y_{i,q}$ denotes one of the three market-quality outcomes for stock i in quarter q : Amihud illiquidity, realized volatility, or price informativeness. $IndexOwn_{i,q}$ denotes the fraction of the firm's shares held by index funds. $X'_{i,q}$ is the vector of control

variables, including $\ln(\text{MarketCap}_{i,q})$ and $\ln(\text{DollarVolume}_{i,q})$, which capture firm size and trading activity. The terms α_i and λ_q denote firm and quarter fixed effects, respectively, while $\epsilon_{i,q}$ is the regression error term.

The coefficient of interest is β , which captures the association between index ownership and each outcome. Because the specification includes firm and time fixed effects, β is identified from within-firm variation over time relative to common trends. This design reduces confounding from time-invariant firm heterogeneity and broad market shifts. Nevertheless, the estimates remain correlational because ownership may respond endogenously to market quality.

1.3.2 VARIABLE CONSTRUCTION

For stock i in quarter q , $\text{IndexOwn}_{i,q} = \frac{\text{IndShares}_{i,q}}{\text{Shrout}_{i,q}}$ is defined as the fraction of a firm's shares held by index funds at quarter end. Here, $\text{IndShares}_{i,q}$ denotes the number of shares held by identified index mutual funds and ETFs, while $\text{Shrout}_{i,q}$ denotes the firm's total shares outstanding. This variable captures the intensity of passive ownership in the shareholder base. Economically, it measures the extent to which a firm is exposed to index-related investment flows and passive ownership. It is the primary explanatory variable in the baseline regressions. Higher values indicate greater exposure to indexing-related trading and information channels. Since $\text{IndexOwn}_{i,q}$ is constructed from reported holdings of index mutual funds and ETFs, it may not fully capture passive investment conducted through other institutional vehicles or channels.

$$\text{Amihud}_{i,q} = \frac{1}{n} \sum_{t=1}^n \frac{|r_{i,t}|}{P_{i,t} * v_{i,t}}$$

Amihud illiquidity is constructed from daily data as the average of absolute returns divided by daily dollar volume within each quarter, where t denotes a trading-day

index within quarter q , $r_{i,t}$ is stock i 's daily return, $P_{i,t}v_{i,t}$ is its daily dollar trading volume, and n is the number of valid trading days. The measure captures the price impact per dollar traded: higher values mean that relatively small amounts of trading are associated with larger price movements, indicating lower liquidity and higher trading costs.

$$\text{Volatility}_{i,q} = \left(\sqrt{\frac{1}{N_q - 1} \sum_{d=1}^{N_q} (r_{i,d,q} - \bar{r}_{i,q})^2} \right) \times \sqrt{N_q}$$

Volatility is computed as the standard deviation of daily returns within each quarter, yielding a realized volatility measure, where d denotes a trading-day index in quarter q , $r_{i,d,q}$ is stock i 's daily return, $\bar{r}_{i,q}$ its mean daily return, and N_q the number of valid trading days. Both variables are first measured at daily frequency and then aggregated to the quarter to capture high-frequency liquidity variation while reducing idiosyncratic noise. This aggregation preserves temporal dynamics relevant for panel estimation and aligns the measures with quarter-end ownership data.

Price informativeness is proxied by a synchronicity-based measure defined as $\ln \left(\frac{1-R_{i,q}^2}{R_{i,q}^2} \right)$, where R^2 is obtained from a daily Fama–French three-factor regression. Intuitively, R^2 measures how much of a stock's daily price movement can be explained by broad market factors. Because the transformation is decreasing in R^2 , higher values indicate that the stock moves less closely with common factors and that its price contains relatively more firm-specific information. The regression is specified as follows:

$$r_{it} - r_{ft} = \alpha_{it} + \beta_1 (r_{Mt} - r_{ft}) + \beta_2 SMB_t + \beta_3 HML_t + \epsilon_{it}$$

Here, r_{it} denotes stock i 's daily return, r_{ft} the daily risk-free rate, and r_{Mt} the daily market return. Thus, $r_{it} - r_{ft}$ represents the stock's excess return, while $r_{Mt} - r_{ft}$

represents the market risk premium. The variables SMB_t and HML_t denote the size and value factors, respectively.

The three-factor model is used as the baseline because the objective is to measure stock-level comovement with broad common factors rather than to estimate a complete asset-pricing model. It captures the market, size, and value dimensions of common return variation while remaining parsimonious for quarterly firm-level estimation. Although the five-factor model adds profitability and investment factors, the resulting synchronicity measures are similar to those from the three-factor specification, with limited incremental benefit in this setting.²

This approach is adopted because synchronicity captures the degree to which firm returns comove with common factors, a feature that reflects the incorporation of market-wide information into prices. Daily excess returns $r_{it} - r_{ft}$ are regressed on the market excess return, SMB, and HML within each quarter to compute firm-level R^2 , thereby retaining high-frequency information before aggregation. The quarterly transformation aggregates daily informativeness in a way that preserves economically meaningful variation across firms and time. The logarithmic transform maps the bounded R^2 into an unbounded scale, improves comparability across firms, and yields coefficients that are more linear in interpretation.

Control variables include $\ln MarketCap_{i,q}$ and $\ln DollarVolume_{i,q}$ log of firm size and dollar trading volume, both computed from CRSP data and aggregated to the quarter. Firm size captures persistent differences in information environments, dislo-

²As a specification check, I compare the R^2 values from the Fama–French 3-factor and 5-factor models for the 100 largest eligible U.S. common stocks at the beginning of 2015. Across 3,721 stock-quarter observations from 2015Q1 through 2024Q4, the mean R^2 is 0.422 under the 3-factor model and 0.477 under the 5-factor model, with a correlation of 0.963 between the two measures. These modest differences indicate that adding the profitability and investment factors provides no substantial incremental gain in this setting, supporting the use of the 3-factor model as the baseline specification.

sure intensity, and investor attention that are correlated with both ownership structure and market quality. Dollar volume proxies for trading activity and liquidity provision, absorbing variation in market participation that could otherwise be attributed to index ownership. Including these controls reduces omitted-variable bias by isolating the partial association between index ownership and the outcome variables. The controls are standard in the market-quality literature (etc, Chordia, Roll, and Subrahmanyam, 2008; Chung and Hrazdil, 2010).

Table 1.1: Panel Regressions with Firm and Time Fixed Effects: Detailed Output

	Amihud illiquidity	Volatility	Price informativeness
Index ownership	0.0613*** (0.0201)	-0.0560** (0.0233)	-3.3018*** (0.2384)
Firm size	0.0084*** (0.0016)	-0.0984*** (0.0031)	-0.3406*** (0.0269)
Dollar volume	-0.0222*** (0.0017)	0.0748*** (0.0024)	0.1453*** (0.0148)
Observations	241,199	240,515	239,758
Entities	8,877	8,858	8,855
Time periods	100	100	100
Firm FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Two-way clustered SE	Yes	Yes	Yes

Notes: This table reports panel fixed-effects regressions with two-way clustered standard errors by firm and date. Standard errors are reported underneath coefficients in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The detailed output preserves the model-level counts and specification flags used in the baseline panel regressions.

Table 1.1 reports the baseline coefficient estimates on index ownership for each outcome. The estimates are obtained from the two-way fixed-effects specification with two-way clustered standard errors, using the ownership-covered subsample. Each column is interpreted as a partial correlation between index ownership and the relevant

market-quality proxy. The paragraphs below summarize the magnitude, sign, and statistical precision for each variable.

Because $IndexOwn_{i,q}$ is measured as a fraction of shares outstanding, β represents the change in the outcome associated with a one-unit increase in ownership, equivalent to 100 percentage points. Since ownership changes are typically smaller than one unit, the coefficient is more informatively interpreted after rescaling it to empirically relevant changes. For example, a one-percentage-point increase corresponds to $\Delta IndexOwn = 0.01$ and changes the outcome by 0.01β . The discussion below therefore evaluates coefficient magnitudes using ownership changes observed in the sample and considers their statistical precision separately.

To express the estimates in economically interpretable units, the analysis considers both a one-percentage-point increase in index ownership and an interquartile-range increase. The one-percentage-point change represents a marginal change in ownership, while the interquartile-range increase reflects the observed movement from the 25th percentile of ownership, 2.6%, to the 75th percentile, 9.5%. This corresponds to an ownership change of 0.069, so the implied change in any outcome is calculated as 0.069β . The interquartile-range benchmark captures a substantial but empirically observed change in ownership and provides a useful basis for evaluating economic magnitude.

For Amihud illiquidity, the coefficient on index ownership is positive and statistically significant, with $\beta = 0.0613$ and a 95% confidence interval of $[0.0219, 0.1008]$. This estimate implies that higher index ownership is associated with higher illiquidity after conditioning on fixed effects and controls. The magnitude is economically meaningful because illiquidity exhibits a heavy right tail, so modest shifts in the mean can correspond to sizable changes in price impact for the most illiquid observations. In addition, the Amihud measure scales returns by dollar volume, which makes it par-

ticularly sensitive to changes in trading conditions during periods of thin liquidity. The result is significant at the 1% level, indicating a robust association in the baseline design.

For volatility, the coefficient on index ownership is negative and statistically significant, with $\beta = -0.0560$ and a 95% confidence interval of $[-0.1018, -0.0103]$. This estimate suggests that higher index ownership is associated with lower realized volatility in the baseline specification. The sign contrasts with concerns that passive ownership mechanically amplifies volatility, which motivates further regime and robustness analysis. The estimate is significant at the 5% level and remains stable within the ownership-covered sample.

For price informativeness, the coefficient is large, negative, and highly significant, with $\beta = -3.3018$ and a 95% confidence interval of $[-3.7691, -2.8345]$. This implies that higher index ownership is associated with lower values of the informativeness proxy in the baseline panel. The magnitude is sizable relative to the cross-sectional dispersion of the proxy, indicating a strong negative association. This result is statistically significant at the 1% level and warrants additional scrutiny of measurement and identification in later chapters.

Rescaling the baseline coefficients to the observed interquartile-range increase in ownership of 0.069 implies an increase in Amihud illiquidity of approximately 0.0042, a reduction in volatility of approximately 0.0039, and a decline in the price-informativeness proxy of approximately 0.228. The illiquidity change is meaningful relative to the sample median of 0.0128, although the comparison is affected by the measure's substantial right skew. The volatility change represents approximately 2%–3% of its mean or median level, suggesting a more modest relative effect. The decline in the informativeness proxy corresponds to roughly 10% of its interquartile range, indicating a nontrivial shift in the information dimension. These comparisons suggest

that the baseline associations are economically relevant, although their magnitudes differ across outcomes and should not be interpreted as causal effects.

Control-variable coefficients vary across outcomes and should be interpreted within the fixed-effects specification rather than against unconditional cross-sectional expectations. Firm size is positively associated with Amihud illiquidity, contrary to the conventional expectation that larger firms are more liquid. Because this coefficient is estimated conditional on dollar volume and firm fixed effects, it should not be interpreted as evidence that larger firms are unconditionally less liquid; rather, it captures the remaining association between firm size and illiquidity after accounting for trading activity. Firm size is negatively associated with volatility and the informativeness proxy. Dollar volume is negatively associated with illiquidity and positively associated with both volatility and informativeness.

The consistency of control coefficients across outcomes suggests that the model captures stable cross-sectional patterns. However, these controls do not resolve the endogeneity of index ownership. The baseline results therefore serve as descriptive benchmarks rather than causal estimates. Later identification strategies will be required to isolate causal effects of index ownership on market quality.

The baseline panel regressions establish three principal results: index ownership is positively associated with illiquidity, negatively associated with volatility, and negatively associated with the price informativeness proxy. These patterns are estimated within a two-way fixed-effects framework and provide a descriptive benchmark across outcomes. At the same time, the mixed signs suggest that indexing may operate through multiple channels rather than a single monotonic mechanism. This motivates a deeper examination of identification and functional form in the chapters that follow.

A central limitation of the baseline results is endogeneity, because index ownership is likely jointly determined with market quality and may respond to unobserved shocks to liquidity, information, or volatility. Such feedback can bias fixed-effects estimates even after controlling for firm and time effects. The next section therefore implements instrumental-variable regressions to isolate variation in index ownership that is plausibly exogenous. Beyond endogeneity, the mixed signs also raise the possibility that effects are nonlinear, with different responses at low versus high ownership intensity. Subsequent sections accordingly examine nonlinear and nonparametric specifications to test whether the baseline correlations vary across the ownership distribution.

1.4 CAUSALITY ANALYSIS

1.4.1 IDENTIFICATION

This chapter estimates the causal effect of index ownership on market-quality outcomes using instrumental variables. The objective is to address endogeneity that arises when ownership responds to unobserved shocks to liquidity, volatility, or information, or when these outcomes influence index inclusion probabilities. A two-stage least squares design is therefore used to isolate exogenous variation in index ownership. The IV estimates complement the baseline fixed-effects results by strengthening causal interpretation.

The identification strategy relies on the structure of the Russell index family. The Russell 3000 is a broad U.S. equity benchmark consisting of the 3,000 largest firms ranked by market capitalization. It is divided into the Russell 1000, which contains the largest 1,000 firms and represents large-cap stocks, and the Russell 2000, which contains the next 2,000 firms and represents small-cap stocks. Because the indexes are value weighted, index assignment affects the demand from funds that track them. In

particular, firms near the bottom of the Russell 1000 receive relatively small weights in that index, whereas similarly sized firms near the top of the Russell 2000 receive larger weights in the small-cap index.

These institutional features provide quasi-exogenous variation in index ownership. Rather than conducting an event study of annual reconstitutions, this analysis uses mechanical, pre-announced market-capitalization rankings to construct cutoff-based instruments. At the Russell 1000 cutoff, firms switch between the Russell 1000 and Russell 2000 while remaining in the Russell 3000. Thus, firms ranked around 1,000 are similar in size but receive different index assignments and index weights. At the Russell 2000 cutoff, which lies between ranks 3,000 and 3,001, firms just inside the Russell 2000 are compared with firms just outside the Russell 3000. This cutoff captures the discontinuity between Russell 2000 membership and nonmembership in the Russell indexes. Identification relies on local comparability around each cutoff, while differences in first-stage strength across the two cutoffs reflect an empirical distinction between the designs. This approach follows prior research on passive ownership and index reconstitution (Ben-David et al., 2018; Chang, Hong, and Liskovich, 2015; Appel, Gormley, and Keim, 2016; Crane et al., 2016).

The instruments satisfy the two conditions required for causal identification: relevance and the exclusion restriction. Relevance holds because the cutoff assignment generates a meaningful change in index ownership. Formally, the relevance condition requires a nonzero covariance between the cutoff indicator and index ownership:

$$\text{Cov}(Z_{i,q}, \text{IndexOwn}_{i,q}) \neq 0$$

The exclusion restriction is assumed to hold because the cutoff assignment affects market-quality outcomes only through its effect on index ownership. Formally, the

instrument must be orthogonal to the error term:

$$Z_{i,q} \perp \varepsilon_{i,q}$$

Firms immediately around each cutoff are similar in pre-reconstitution fundamentals because index assignment is determined by a mechanical market-capitalization ranking. The cutoff designs capture two types of transitions: reassignment between the Russell 1000 and Russell 2000 at the Russell 1000 cutoff, and entry into or exit from the Russell 2000 at the Russell 2000 cutoff. These transitions generate variation in index ownership, and the exclusion restriction requires that they affect market-quality outcomes through this ownership channel rather than through a separate discontinuous change at the cutoff.

1.4.2 2SLS REGRESSION SETUP

First Stage:

$$\begin{aligned} IndexOwn_{i,q} &= \pi_0 + \pi_1 Z_{i,q} + \gamma X'_{i,q} + \alpha_i + \lambda_q + u_{i,q} \\ \text{where} \\ Z_{i,q}^{(c)} &= 1 \{ \text{Rank}_{i,q} \leq c \} , \\ c &= \text{cutoff rank, } c \in \{1,000^1, 3,000^2\} \end{aligned} \tag{1.2}$$

Second Stage:

$$Y_{i,t} = \beta_0 + \beta_1 \widehat{IndexOwn}_{i,q} + \gamma X'_{i,q} + \alpha_i + \lambda_q + \epsilon_{i,q} \tag{1.3}$$

Table 1.2 reports first-stage diagnostics for the global IV specifications at the Russell 1000 and Russell 2000 cutoffs. Using the full sample with a sharp indicator

¹At the Russell 1000 cutoff, $c = 1,000$, separating ranks $[1, 1,000]$, assigned to the Russell 1000, from ranks $[1,001, 3,000]$, assigned to the Russell 2000.

²At the Russell 2000 cutoff, $c = 3,000$, separating ranks $[1,001, 3,000]$, assigned to the Russell 2000, from ranks $[3,001, \infty)$, outside the Russell 3000.

yields strong first-stage F statistics: approximately 28 at the Russell 1000 cutoff and above 128 at the Russell 2000 cutoff. These values indicate that the instruments are strong in the global specifications, producing precise fitted ownership variation. First-stage F-statistics assess whether the cutoff assignments generate sufficient variation in index ownership for IV estimation. Both cutoffs satisfy this relevance condition, although the association is stronger at the Russell 2000 cutoff. The larger F-statistic indicates stronger first-stage relevance and reduces concerns about weak-instrument bias, but does not necessarily imply a larger ownership discontinuity.

Table 1.2: First-Stage F Statistics for Global IV Cutoffs

	Russell 1000	Russell 2000
Amihud illiquidity	27.7911***	131.1626***
Volatility	27.3002***	128.9754***
Price informativeness	28.4864***	131.1246***
Observations	235,544	235,544
Firms	8,627	8,627
Dates	99	99
Clusters	8,627	8,627
Controls	Yes	Yes
Fixed effects	cusip FE + date FE	cusip FE + date FE
SE type	firm clustered	firm clustered

Notes: This table reports first-stage relevance diagnostics for the global cutoff-IV design. Entries are first-stage F statistics for the excluded cutoff indicator in the first-stage regression of instrumented ownership. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The sample is the global specification with no bandwidth restriction, estimated separately at the Russell 1000 and Russell 2000 cutoffs.

Local RD-IV estimates provide a credibility check but exhibit weak first stages in narrow bandwidths. Within a ± 300 -rank window around the 2000 cutoff and a ± 150 window around the 1000 cutoff, first-stage F statistics fall well below conventional thresholds. This weakness is expected because the ownership discontinuity is smaller in very narrow windows and noise dominates local variation. Accordingly, detailed

local RD-IV results are reported as robustness check in the Appendix section, where the trade-off between identification purity and statistical power is examined explicitly.

The contrast between local and global IV results reflects a standard trade-off in reconstitution designs. Local RD-IV estimates are more credible with respect to the exclusion restriction but suffer from weak first stages due to limited ownership discontinuities near the threshold. Global specifications yield strong instruments and precise estimates but rely on the stronger assumption that potential outcomes vary smoothly across the full rank distribution. Following common practice, the global estimates are treated as primary results, while local estimates serve as robustness checks.

Table 1.3: IV Regression Results: Russell 1000

	Russell 1000
Amihud illiquidity	0.2319 (0.1450)
Volatility	-1.4850*** (0.4090)
Price informativeness	-5.8808 (4.0159)
Observations	235544
Firms	8627
Dates	99
Clusters	8627
Endogenous variable	<i>ind_own</i>
Instrument	<i>z_rank_le_cutoff</i>
Controls	<i>mktcap; c_dollar_vol; r_tno; rank_minus_cutoff</i>
Fixed effects	cusip FE + date FE
SE type	firm clustered

Notes: This table reports second-stage IV estimates for the global cutoff-IV design. Standard errors are reported underneath coefficients in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

The sample is the global specification with no bandwidth restriction, estimated at the Russell 1000 cutoff.

Table 1.3 summarizes the global IV estimates at the Russell 1000 cutoff. Index ownership is positively associated with Amihud illiquidity ($\beta = 0.2319$, $p \approx 0.11$), negatively associated with volatility ($\beta = -1.4850$, $p < 0.001$), and negatively associated with price informativeness ($\beta = -5.8808$, $p \approx 0.14$).

In outcome units, a one-percentage-point increase in index ownership corresponds to a 0.0023 increase in Amihud illiquidity, a 0.0149 reduction in realized volatility, and a 0.0588 reduction in the price-informativeness proxy. The volatility estimate is economically meaningful relative to the sample median volatility of 0.150, representing a reduction of approximately 10%, and is statistically precise. By contrast, the illiquidity and informativeness estimates are statistically imprecise, so their point estimates should not be interpreted as reliable evidence of economically meaningful changes at this cutoff.

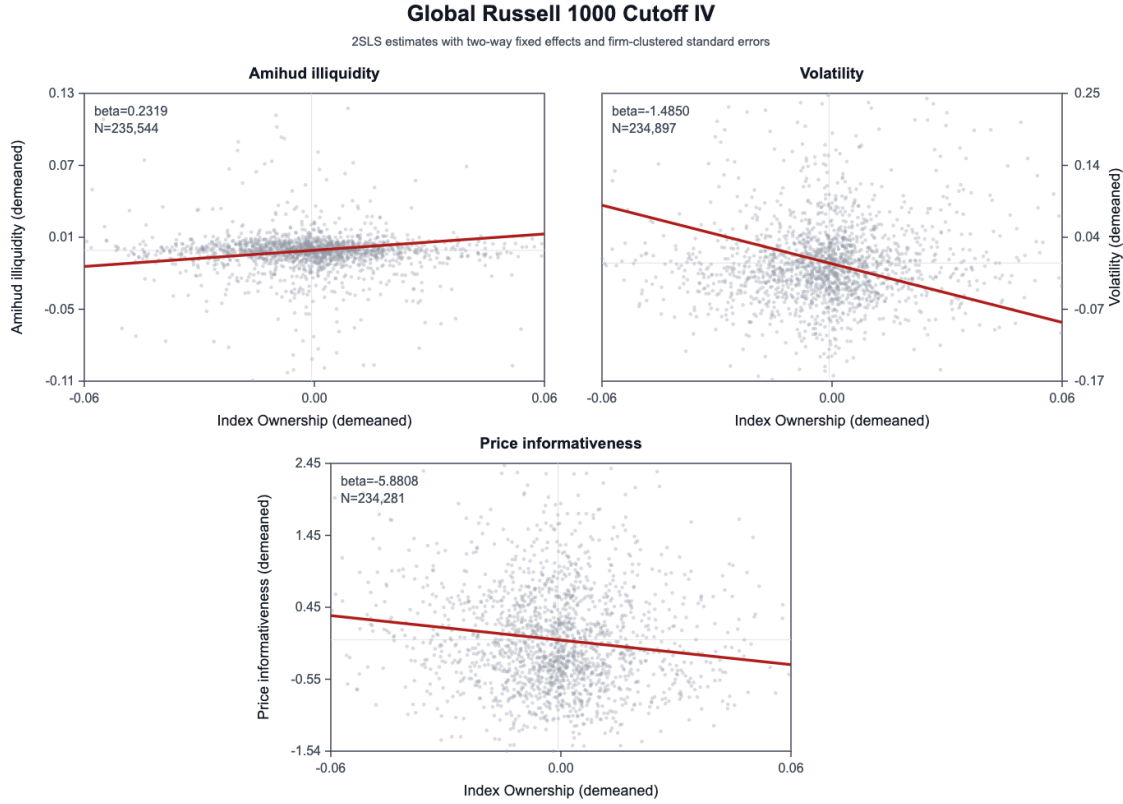


Figure 1.2: 2SLS Regression Results of R1000

Notes: This figure plots fitted IV relationships between index ownership and each market-quality outcome using the global Russell cutoff 1000 specification.

Figure 1.2 presents the IV results at the Russell 1000 cutoff for all three market quality measures. For Amihud illiquidity, the upward line would imply a deteriorated liquidity in the more passive market. However, the estimate is not statistically significant. Therefore, the Russell 1000 IV results do not support a reliable liquidity conclusion. The volatility result is materially different. The fitted line in the volatility panel slopes downward in a clear and economically meaningful way. This indicates that higher index-related ownership reduces idiosyncratic volatility for firms near the large-cap boundary. For the price informativeness proxy, the estimated coefficient is also negative. The sign is directionally consistent with the baseline concern that passive ownership may weaken firm-specific price discovery, yet the estimate is too

statistically imprecise to support a meaningful causal conclusion. Taken together, the cutoff-1000 plot points primarily to lower volatility, while the illiquidity and informativeness estimates are directionally consistent but less precise. The main implication of the Russell 1000 specification is therefore a volatility-reduction channel.

Table 1.4: IV Regression Results: Russell 2000

	Russell 2000
Amihud illiquidity	-0.2849*** (0.0776)
Volatility	-0.1767 (0.1263)
Price informativeness	-12.0868*** (1.7568)
Observations	235544
Firms	8627
Dates	99
Clusters	8627
Instrument	<i>z_rank_le_cutoff</i>
Controls	<i>mktcap; c_dollar_vol; r_tno; rank_minus_cutoff</i>
Fixed effects	cusip FE + date FE
SE type	firm clustered

Notes: This table reports second-stage IV estimates for the global cutoff-IV design. Standard errors are reported underneath coefficients in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The sample is the global specification with no bandwidth restriction, estimated at the Russell 2000 cutoff.

The global cutoff-2000 specification is substantially stronger from an identification standpoint. Its first-stage F-statistics are around 129 to 131, indicating a much more powerful relationship between index assignment and index ownership. This stronger first stage gives the second-stage coefficients greater credibility and makes the visual patterns in the fitted-line plots more informative.

Table 1.4 reports the global IV estimates at the Russell 2000 cutoff. Index ownership is associated with lower illiquidity ($\beta = -0.2849$, $p < 0.001$), indicating a sizable

liquidity improvement for firms entering the small-cap index. The price informativeness coefficient is large and negative ($\beta = -12.0868$, $p < 10^{-11}$), suggesting that increased passive ownership is linked to lower values of the synchronicity-based proxy. The volatility estimate is negative but statistically weak ($\beta = -0.1767$, $p \approx 0.16$), implying that volatility effects are less precisely identified at this cutoff. Taken together, the 2000-cutoff results point to a strong liquidity channel and a pronounced informativeness response, while volatility effects remain ambiguous.

A one-percentage-point increase in index ownership corresponds to a 0.0028 reduction in Amihud illiquidity, a 0.0018 reduction in realized volatility, and a 0.1209 reduction in the price-informativeness proxy. The illiquidity estimate represents approximately 22% of the sample median illiquidity measure of 0.0128, indicating a potentially meaningful improvement in trading conditions. The informativeness estimate indicates a decline in firm-specific information incorporated into prices, although its magnitude is smaller relative to the overall dispersion of the proxy. The volatility effect is economically small and statistically imprecise. Overall, the Russell 2000 results suggest a trade-off between improved liquidity and weaker price informativeness.

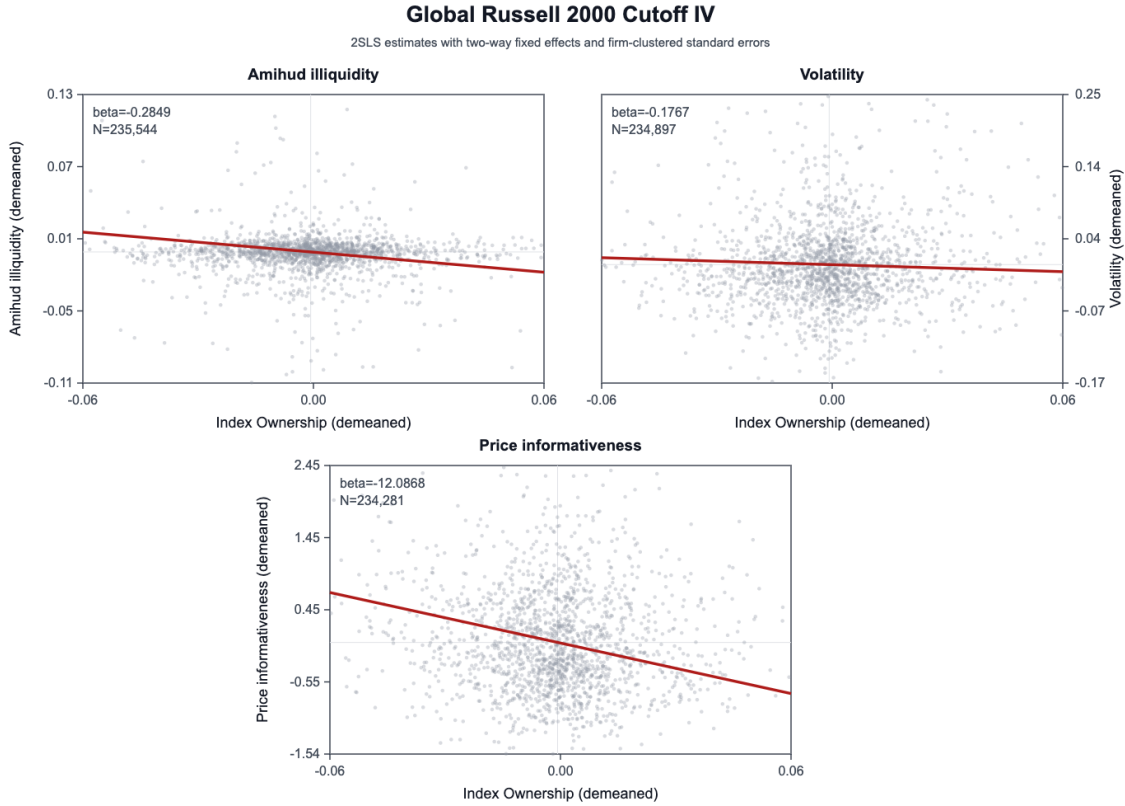


Figure 1.3: 2SLS Regression Results of R2000

Notes: This figure plots fitted IV relationships between index ownership and each market-quality outcome using the preferred global Russell cutoff 2000 specification. .

In Figure 1.3, the fitted line in the illiquidity panel slopes downward in a manner that matches the regression result, showing the liquidity is improved in a market with higher indexing investment. Accordingly, the cutoff-2000 IV evidence suggests that passive ownership can improve trading conditions for smaller firms. The fitted line in the price-informativeness panel is clearly downward sloping and is visually the most persuasive panel among the cutoff-2000 plot. This result supports the view that passive ownership weakens firm-specific information production or incorporation into prices among smaller firms near the Russell 2000 boundary. In this specification, the information environment appears to be the main cost associated with index-driven ownership changes.

The fitted line in the volatility panel also slopes downward, but less decisively than the lines for illiquidity and informativeness. It is not statistically significant and the evidence is not precise enough to support a firm conclusion. Therefore, the overall interpretation of the cutoff-2000 plot is that higher index ownership improves liquidity while simultaneously reducing the informativeness proxy, with the volatility channel remaining comparatively weak.

1.4.3 WHY CUTOFFS PRODUCE DIFFERENT EFFECTS

There is an interesting observation that the two global cutoff specifications imply materially different causal effects. The estimated effects differ in both economic channel and statistical precision. At the Russell 1000 cutoff, the only robust result is the reduced-volatility channel, while the estimated effects on liquidity and informativeness are statistically weak. At the Russell 2000 cutoff, the volatility effect fades, but the liquidity and informativeness effects become prominent and statistically significant. This pattern indicates that the effect of passive ownership is not stable across the size distribution. Instead, the dominant adjustment margin depends on the type of firms exposed to the ownership shock.

Although the liquidity effect in the Russell 1000 sample is not statistically significant, this sign reversal is economically suggestive. Around the Russell 1000 boundary, the illiquidity coefficient is positive but imprecise, whereas around the Russell 2000 boundary it is negative and strongly significant. The Russell 1000 estimate should not be overstated because it lacks statistical precision, yet the sign reversal is still economically suggestive. It implies that passive ownership may interact differently with pre-existing trading environments in large-cap and small-cap stocks. In large-cap firms, additional passive ownership appears unable to generate clear liquidity

gains, while in smaller firms it is associated with a measurable improvement in trading conditions.

The divergent patterns around the Russell 1000 and Russell 2000 cutoffs are consistent with systematic differences in firm characteristics and market structure. Firms near the Russell 1000 threshold are relatively large, trade in deeper markets, and receive broader attention from investors and intermediaries. In that environment, baseline liquidity and information production are already comparatively strong, so an increase in passive ownership is less likely to generate sizable marginal improvements in transaction costs or price efficiency. Instead, the effect of indexing is more likely to appear through trading-related channels, such as correlated fund flows and changes in return dynamics. Firms around the Russell 2000 cutoff, by contrast, are smaller, less liquid, and subject to greater informational frictions, so the same increase in index ownership represents a more meaningful expansion in market participation.

This difference in market structure helps explain why the Russell 2000 specification produces stronger effects on liquidity and the price informativeness proxy. When the initial investor base is thinner, passive inflows can increase trading activity and reduce trading frictions, so the ownership shock has a larger effect on market quality. By contrast, when those frictions are already limited, as for firms near the Russell 1000 cutoff, the incremental effect of passive ownership is correspondingly smaller and less likely to appear in standard liquidity measures. The contrast between the two cutoffs therefore points to heterogeneity in the causal effect of indexing rather than inconsistency in the empirical evidence.

Two mechanisms are particularly relevant. First, passive inflows may serve as a net addition to trading activity among smaller-cap stocks, where the marginal value of new participation is high. Second, substitution between active and passive investors is likely to be more pronounced among firms near the Russell 1000 cutoff, where passive

products are already widely used and active traders continue to play an important role in price discovery. If passive ownership expands mainly by displacing active investors in that segment, the gains in liquidity and information efficiency may be attenuated. In addition, larger firms are more deeply embedded in exchange-traded funds and benchmark-tracking portfolios, which can generate correlated trading flows that increase return comovement and amplify non-fundamental volatility without necessarily improving liquidity or price efficiency.

Differences in treatment intensity and identification further support this interpretation. The discontinuity in index ownership is typically larger around the Russell 2000 cutoff than around the Russell 1000 cutoff, which yields a stronger first stage and more precise second-stage estimates. As a result, the Russell 2000 specification is better positioned to detect how passive ownership affects market quality when the underlying firms are more sensitive to capital flows. The two cutoffs therefore identify effects in distinct local environments: the Russell 1000 captures relatively liquid, information-rich large-cap settings, whereas the Russell 2000 captures smaller-cap settings in which market frictions are more substantial and the ownership shock is more economically meaningful.

Taken together, these findings imply that the causal effect of index ownership is context-dependent rather than uniform across the cross section. The two thresholds should therefore be interpreted as complementary evidence on how indexing operates across different firm size regimes rather than as competing empirical conclusions. This interpretation motivates the next stage of the analysis, which turns to size-sorted panel regressions in order to examine whether similar patterns appear more broadly across the full size distribution.

These findings raise the question of whether the ownership–market-quality relationship is nonlinear across the distribution of index ownership. If marginal effects

differ between low- and high-ownership regimes, linear IV estimates may mask important heterogeneity. The next section therefore examines nonlinear and nonparametric specifications to test whether the causal effects vary with ownership intensity and firm characteristics. This extension clarifies whether the cutoff-specific differences reflect broader nonlinear patterns in the data.

1.5 NONLINEARITY ANALYSIS

1.5.1 QUADRATIC ANALYSIS SET-UP

This section extends the baseline panel and IV analyses by examining whether the relationship between index ownership and market quality is nonlinear. The linear specifications documented systematic effects but also produced mixed signs across outcomes and cutoffs, raising the possibility that marginal effects vary with ownership intensity. A nonlinear framework can reconcile these patterns by allowing the slope to change over the ownership distribution.

Exploring nonlinearity is economically motivated because the mechanisms of indexing need not scale proportionally. Small increases in passive ownership may improve liquidity and information aggregation, while very high ownership could dampen trading incentives or concentrate demand. Such threshold effects would be masked in linear models but are consequential for policy and interpretation. The analysis therefore estimates quadratic specifications and constructs a composite market-quality score to identify an interior optimum.

The baseline nonlinear specification augments the fixed-effects model with a quadratic term:

$$Y_{i,q} = \alpha + \beta_1 \text{IndexOwn}_{i,q} + \beta_2 \text{IndexOwn}_{i,q}^2 + \gamma X'_{i,q} + \alpha_i + \lambda_q + \epsilon_{i,q} \quad (1.4)$$

where the turning point is defined as

$$\text{IndexOwn}^* = -\frac{\beta_1}{2\beta_2}$$

The quadratic equations summarize the estimated relationship through the linear and squared ownership terms. Figure 1.4 translates these estimates into fitted curves over the observed ownership support, normalized relative to the median ownership level. The panels show the shape of the estimated relationship—whether it is monotonic or turns within the support—rather than the individual coefficients or their standard errors. The discussion below therefore focuses on the direction, curvature, and implied turning point for each outcome.

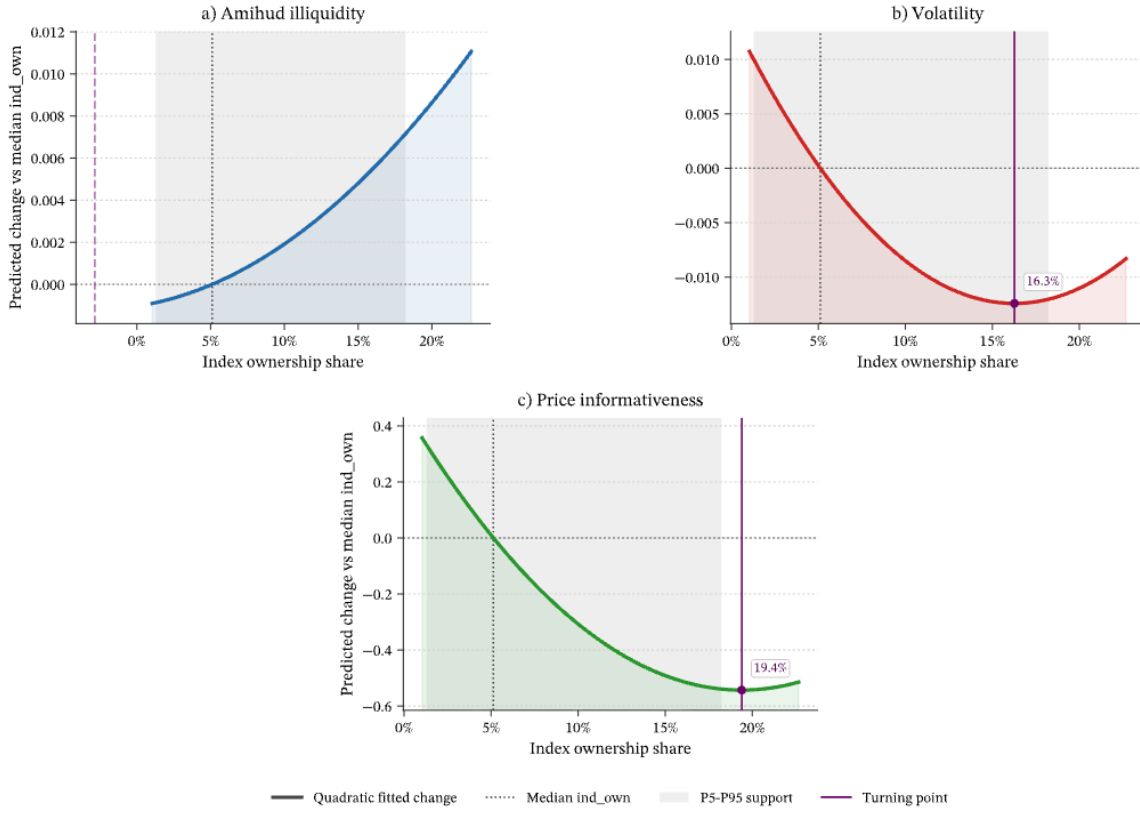


Figure 1.4: Quadratic Panel Regression Results

Notes: This figure shows the non-IV quadratic relationship between index ownership and the three market-quality outcomes. Each panel traces the fitted nonlinear effect over the support of index ownership, illustrating whether the association between indexing and market quality is monotonic or exhibits an interior optimum.

Volatility exhibits a clear U-shape: marginal increases in index ownership reduce volatility at low levels, but the benefit diminishes and eventually reverses at higher ownership. Price informativeness displays a similar U-shape, implying that moderate indexing improves informativeness while very high ownership may dilute firm-specific signals. Amihud illiquidity shows weaker curvature, with the quadratic term significant but the turning point outside the observed support, suggesting a less stable liquidity response in the reduced-form setting.

Figure 1.4 also provides an indication of economic magnitude because it expresses estimated changes relative to median ownership. Across the observed ownership support, Amihud illiquidity increases by approximately 0.012, a change comparable to its sample median of 0.0128. Volatility declines by roughly 0.02 before reaching its minimum at approximately 16.3% ownership, representing a moderate change relative to the sample median volatility of 0.150. The price-informativeness proxy declines by approximately 0.8 to 0.9 units before reaching its minimum near 19.4% ownership, a nontrivial movement relative to its interquartile range. These changes suggest that the nonlinear patterns are economically relevant, although their interpretation remains descriptive because the reduced-form specifications do not address ownership endogeneity.

These patterns imply that indexing is most beneficial at low-to-moderate levels, where passive participation can deepen trading and enhance information aggregation without displacing active monitoring. As ownership rises, marginal benefits taper and may reverse, consistent with reduced incentives for information production or concentrated demand elasticity.

High levels of index ownership may be undesirable because they can reduce the incentives for costly information production and amplify demand inelasticity. When passive ownership is dominant, fewer investors trade on firm-specific signals, which can weaken price discovery and reduce informational trading depth. In addition, index-linked demand can concentrate order flow and increase co-movement, potentially elevating systemic sensitivity in stressed periods. These mechanisms provide an economic rationale for the observed reversal in marginal benefits at higher ownership levels.

1.5.2 COMPOSITE MARKET-QUALITY SCORE

In empirical economics, summary indices are commonly used when a treatment or policy may affect several related outcomes. Standardizing each outcome as a z-score places measures with different units on a common scale, allowing the aggregate index to reflect relative rather than raw magnitudes. Kling, Liebman, and Katz (2007) use standardized outcomes to construct a summary index, while Anderson (2008) develops an alternative inverse-covariance-weighted approach. The finance literature likewise treats market quality as a multidimensional concept involving liquidity, volatility, and price discovery or efficiency, rather than as a single universally accepted measure (O'Hara and Ye, 2011). Building on these practices, the analysis constructs a composite market-quality score in which higher values represent a more favorable joint outcome across the underlying dimensions, while the individual measures remain available for interpreting specific channels.

Let $\mathcal{G} = \{x_1, \dots, x_G\}$ denote the ownership grid over the observed support, and let $\widehat{Y}_j(x_g)$ denote the fitted value for outcome j at ownership level x_g . For each outcome, the fitted values are standardized across the ownership grid:

$$z_j(x_g) = \frac{\widehat{Y}_j(x_g) - \overline{\widehat{Y}_j}}{s_j}, \quad (1.5)$$

where j indexes the three outcomes, $g = 1, \dots, G$ indexes the ownership-grid points, and $\overline{\widehat{Y}_j}$ and s_j are the mean and standard deviation of the fitted values for outcome j over $\mathcal{G}$. The composite score is then calculated as

$$\text{Score}(x_g) = -z_{\text{amihud}}(x_g) - z_{\text{volatility}}(x_g) + z_{\text{price_info}}(x_g). \quad (1.6)$$

Thus, the score is constructed from the fitted outcome curves rather than from the raw variables or regression coefficients. Standardization removes differences in measurement units and places the three outcomes on a common scale before aggregation.

The composite score assigns equal absolute weights to the three standardized outcomes. This choice is deliberate: after z-score standardization, each component is measured in standard-deviation units, so no outcome dominates the index because of its original scale or dispersion. Equal weighting also provides a transparent and replicable benchmark when theory does not provide a defensible basis for ranking the relative importance of liquidity, volatility, and price informativeness. The signs are chosen only to orient the measures consistently, with higher values representing better market quality. Accordingly, the score should be interpreted as a descriptive summary measure rather than a structural welfare index; it implies that a one-standard-deviation improvement in one dimension offsets a one-standard-deviation deterioration in another. Dividing the sum by three would only rescale the index and would not change the location of its maximum.

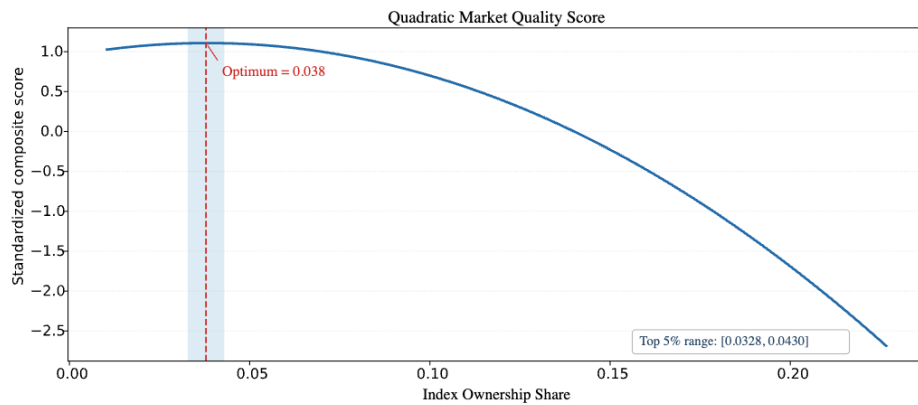


Figure 1.5: Quadratic Analysis - Market Score

Notes: This figure reports the composite market-quality score implied by the non-IV quadratic specification. Higher values indicate a more favorable combination of higher liquidity, lower volatility, and higher price informativeness. The highlighted region marks the range of index ownership associated with the highest composite market quality.

Figure 1.5 plots the score over the ownership grid and highlights the ownership interval that falls in the top 5% of scores. The score peaks at $\text{ind_own} \approx 0.0379$, with a top 5% range of approximately $[0.0328, 0.0430]$. This range implies that low-to-moderate indexing balances the liquidity and volatility benefits of passive participation with the information costs of high ownership. Low-to-moderate indexing level seems to be optimal for market quality under the composite criterion. The narrow high-quality range indicates that marginal effects change quickly around the optimum, so small deviations can reduce overall market quality. This result translates the nonlinear estimates into a concrete reference for policy-relevant statement about the level of indexing most consistent with market-quality improvement.

1.5.3 QUADRATIC IV ANALYSIS

The current design is reduced-form and may still reflect endogeneity in index ownership. To assess whether the nonlinear patterns persist under causal identification, the analysis extends the IV framework to a quadratic specification.

The Quadratic IV equation is defined as:

$$Y_{i,q} = \alpha + \beta_1 \widehat{\text{IndexOwn}}_{i,q} + \beta_2 \widehat{\text{IndexOwn}^2}_{i,q} + \gamma X'_{i,q} + \alpha_i + \lambda_q + \epsilon_{i,q} \quad (1.7)$$

where $\widehat{\text{IndexOwn}}_{i,q}$ and $\widehat{\text{IndexOwn}^2}_{i,q}$ are the fitted values from the first-stage regressions. The cutoff indicator is denoted by $Z_{i,q}$, and $R_{i,q}$ denotes the normalized running variable.

The two first-stage projections can be written compactly in matrix form as:

$$\begin{bmatrix} IndexOwn_{i,q} \\ IndexOwn^2_{i,q} \end{bmatrix} = \mathbf{\Pi} \begin{bmatrix} Z_{i,q} \\ Z_{i,q}R_{i,q} \\ Z_{i,q}R^2_{i,q} \end{bmatrix} + \mathbf{\Gamma} \begin{bmatrix} c_firm_size_{i,q} \\ c_dollar_vol_{i,q} \\ R_{i,q} \\ R^2_{i,q} \end{bmatrix} + \boldsymbol{\alpha}_i + \boldsymbol{\lambda}_q + \mathbf{u}_{i,q}, \quad (1.8)$$

The matrix equation represents the first-stage system of reduced-form projections used in the quadratic IV estimation. The endogenous-variable vector contains both the level of index ownership and its squared term because both variables enter the second-stage outcome equation. The instrument vector consists of the Russell cutoff indicator $Z_{i,q}$, together with its interactions with the normalized running variable $R_{i,q}$ and $R^2_{i,q}$.

The running variable measures each firm's distance from the selected Russell cutoff based on its market-capitalization rank. Including $R_{i,q}$ and $R^2_{i,q}$ as controls absorbs smooth rank-related differences in ownership and market-quality outcomes, while the interaction terms allow the cutoff effect to vary with distance from the threshold. Thus, identification comes from the discrete cutoff assignment rather than from rank distance itself. The control vector includes firm size and dollar volume, while firm and date fixed effects absorb persistent firm characteristics and common time shocks. The fitted values from both first-stage projections are then used to estimate the quadratic effect of index ownership on each market-quality outcome.

The preferred IV specification uses the global cutoff at 2000, which provides strong first-stage diagnostics for both `ind_own` and `ind_own`². Figure 1.6 presents the quadratic IV fitted lines under the preferred global Russell 2000 design.

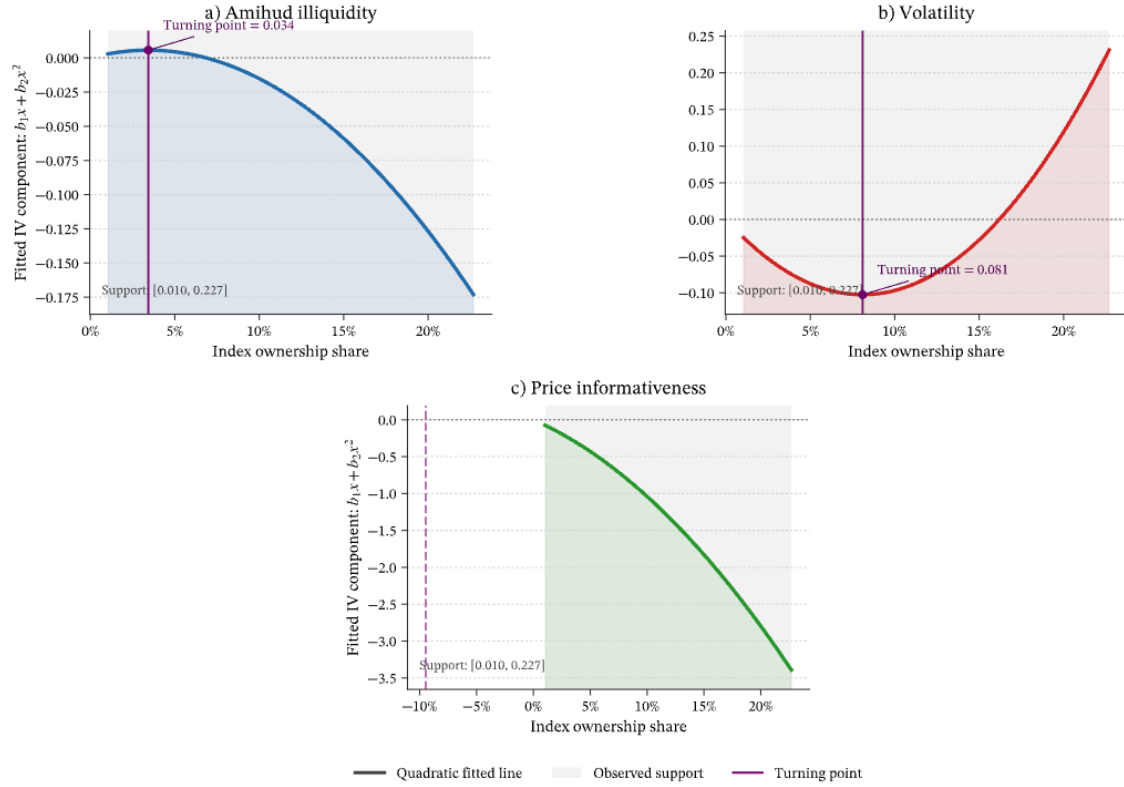


Figure 1.6: Quadratic IV Estimates

Notes: This figure plots the fitted quadratic IV component for each market-quality outcome under the preferred global Russell cutoff 2000 specification. .

The negative values in the price-informativeness panel do not represent negative R^2 . They reflect the fitted quadratic component of the unbounded log synchronicity proxy. Since neither the linear nor quadratic IV coefficient is statistically significant, the curve should be interpreted as an imprecisely estimated pattern rather than evidence of a causal decline in price informativeness.

For volatility, the IV estimates display a clear U-shape, with $\beta_1 = -2.5336$ and $\beta_2 = 15.6492$ and a turning point around $\text{ind_own} \approx 0.0809$ within the support. This pattern implies that low-to-moderate indexing reduces volatility, while higher

ownership levels are associated with diminishing and eventually reversing marginal benefits.

The liquidity and informativeness panels are less decisive. For Amihud illiquidity, the quadratic term is negative and statistically significant ($\beta_2 = -4.8236$), but β_1 is not significant, indicating weaker and less stable curvature. For price informativeness, the nonlinear terms are not statistically significant, suggesting limited IV evidence of curvature in that channel.

The cutoff-1000 quadratic IV design is not retained as a primary specification because the first-stage diagnostics become unstable and not economically meaningful; the corresponding results are reported as a robustness check in the Appendix section. This instability likely reflects weaker ownership discontinuities and a noisier instrument when nonlinear terms are introduced, which reduces power and inflates standard errors. In quadratic IV settings, instruments must be strong enough to identify both the level and the square of ownership, a requirement that the cutoff-1000 design does not satisfy. As a result, the 1000-cutoff estimates are treated as a failed robustness check rather than credible nonlinear evidence.

More broadly, quadratic IV estimation is demanding because it amplifies weak-instrument concerns and sensitivity to specification. Even with a strong first stage for `ind_own`, identification of `ind_own`² can be fragile if the instrument generates limited curvature in fitted values. This limits the precision of higher-order terms and explains why only the 2000-cutoff specification yields interpretable volatility curvature. These constraints motivate cautious interpretation and the use of multiple specifications to triangulate the nonlinear effects.

Despite these limitations, the quadratic IV results are economically meaningful in the channel where identification is strongest. The volatility U-shape implies that the marginal effect of indexing is beneficial at low-to-moderate ownership levels but

becomes less favorable as ownership intensifies, which aligns with theories of reduced information production and demand inelasticity at high passive shares. The lack of robust curvature for price informativeness suggests either weaker causal effects or insufficient power to detect nonlinearities in that channel. Accordingly, the quadratic IV results should be interpreted as evidence of a volatility-specific nonlinear mechanism rather than a universal nonlinear effect across all market-quality dimensions.

This part shows that nonlinearity is a first-order feature of the relationship between index ownership and market quality. In reduced-form quadratic FE models, both volatility and price informativeness display U-shaped patterns, while illiquidity shows weaker curvature with an out-of-support turning point. A composite market-quality score aggregates these curves and indicates an interior optimum at $\text{ind_own} \approx 0.0379$, with a narrow high-quality range around 3.3%–4.3%. These results imply that low-to-moderate indexing tends to improve market quality, while very high ownership can erode marginal benefits.

The IV-based nonlinear analysis refines this conclusion by isolating causal curvature where identification is strongest. Using the cutoff-2000 design, volatility exhibits a robust U-shape with a turning point around $\text{ind_own} \approx 0.0809$, whereas illiquidity curvature is weaker and price informativeness shows no robust nonlinear response. The cutoff-1000 quadratic IV is not retained as a primary specification because its first-stage diagnostics are unstable; those results are reported as robustness checks in the appendix. Overall, the evidence indicates that nonlinear effects are most credible in the volatility channel, while other channels are less stable in IV space.

Taken together, the quadratic reduced-form and IV results show that indexing ownership is associated with market quality in a clearly nonlinear manner, but the evidence also suggests that a single global quadratic restriction is too coarse to summarize the full pattern. The estimated turning points are informative, yet they im-

pose one smooth curvature over the entire support and therefore cannot distinguish whether the relationship changes gradually across the ownership range or shifts more sharply at specific intervals. This limitation is especially relevant because the volatility result is strongest at low-to-moderate ownership levels, while the liquidity and informativeness channels appear weaker and less stable. A spline specification is therefore the natural next step: it preserves a flexible nonlinear structure, allows the slope to vary across segments, and provides a more credible basis for testing whether the apparent interior optimum is stable or only an artifact of the quadratic functional form.

1.5.4 SPLINE ANALYSIS

The spline analysis begins from the same question as the quadratic analysis: whether index ownership impacts market quality, and if so, over what range of ownership the improvement is strongest. The earlier quadratic results suggest interior curvature, but a single global polynomial may be too restrictive to summarize the shape of data. A flexible spline is a more appropriate tool as it allows the slope to vary across the ownership distribution while preserving a smooth fit over the observed support. In other words, the spline specification tests whether the apparent optimum survives once the data are allowed to determine the local slope more freely.

In practice, the effects of index ownership on market functioning are likely to differ across regimes - between low, medium and high levels of ownership. The quadratic form assumes that marginal effects change smoothly and symmetrically around a unique point, which may mask the locally varying dynamics. By allowing curvature to vary across the support, the spline framework provides a piecewise-polynomial approximation of the relationship. It is better suited to trace that tradeoff and to

show whether the estimated interior region is stable across reduced-form and IV specifications.

The spline specification re-estimates the relationship between index ownership and market quality using a natural cubic spline rather than a quadratic polynomial. The spline allows the slope to vary across the ownership distribution, which is useful when the benchmark quadratic suggests an interior optimum but may impose too much global smoothness. To keep the fit economically relevant, ownership is clipped to the pooled $q_{01} - q_{99}$ support³ and rescaled to $[0,1]$. Three interior knots are placed at the 25th, 50th, and 75th percentiles, corresponding to raw values 0.0265, 0.0510, and 0.0952. Standard errors are two-way clustered by firm and date in the reduced-form panel model. The resulting estimates should be read as descriptive evidence on local curvature rather than as a causal design.

Table 1.5: Descriptive Statistics of Index Ownership for Spline Support

Sample	N	Mean	q_{01}	q_{25}	Median	q_{75}	q_{99}	Maximum
Pooled	241,199	6.86%	1.04%	2.65%	5.10%	9.52%	22.67%	95.26%
2000Q4	1,026	1.80%	0.99%	1.28%	1.65%	2.16%	4.02%	36.81%
2005Q4	2,205	2.85%	1.03%	1.91%	2.73%	3.71%	5.59%	11.00%
2010Q4	2,352	5.35%	1.09%	3.46%	5.36%	7.04%	10.97%	31.88%
2015Q4	2,663	7.37%	1.10%	4.38%	7.52%	10.37%	14.86%	30.71%
2020Q4	3,329	10.65%	1.12%	5.97%	10.22%	15.51%	23.30%	55.93%
2023Q4	3,229	11.91%	1.17%	6.46%	11.78%	17.48%	24.62%	91.06%

Notes: This table reports descriptive statistics for quarterly index ownership, expressed as a percentage of firm equity. The pooled row uses all observations with nonmissing ownership, while the snapshot rows report fourth-quarter observations. The pooled q_{01} – q_{99} range defines the common spline support, and the pooled q_{25} , median, and q_{75} values determine the interior knots. Snapshot rows are reported for diagnostic purposes only. Source: Author’s calculations from the quarterly panel.

³Here, q_{01} and q_{99} denote the 1st and 99th percentiles of the pooled index-ownership distribution. These bounds define the central 98 percent of the observed support; observations outside the interval are clipped to the nearest bound to limit the influence of sparse tail observations.

Table 1.5 summarizes the pooled and snapshot ownership distributions used to define the spline support and interior knots. The distribution is right-skewed and shifts upward over time. In the pooled sample, the 1st and 99th percentiles are 1.04% and 22.67%, respectively, while the maximum reaches 95.26%. The median rises from 1.65% in 2000Q4 to 11.78% in 2023Q4, supporting a common pooled support rather than time-varying trimming.

Let $IO_{i,q}$ to be index ownership and k be a knot,

$$Y_{iq} = \beta_1 IO_{iq} + \sum_{k=1}^K \beta_{k+1} (IO_{iq} - \kappa_k)^+ + \gamma X'_{iq} + \alpha_i + \lambda_q + \varepsilon_{iq} \quad (1.9)$$

where

$$(IO_{it} - \kappa_k)^+ = \max(0, IO_{it} - \kappa_k), \quad k = 1, 2, 3, \dots$$

Stabilization is applied because the raw ownership distribution is highly skewed and contains a sparse upper tail. Clipping the series to the pooled $q_{01} - q_{99}$ support and rescaling it to $[0,1]$ keeps the fitted curve focused on the portion of the data where the sample is dense enough to identify meaningful curvature. This also reduces the risk that the spline is driven by a few extreme observations, which can create unstable turning points or misleading tail behavior. In economic terms, stabilization ensures that the estimated shape reflects the relevant interior tradeoff between liquidity, volatility, and price informativeness rather than extrapolation at the extremes.

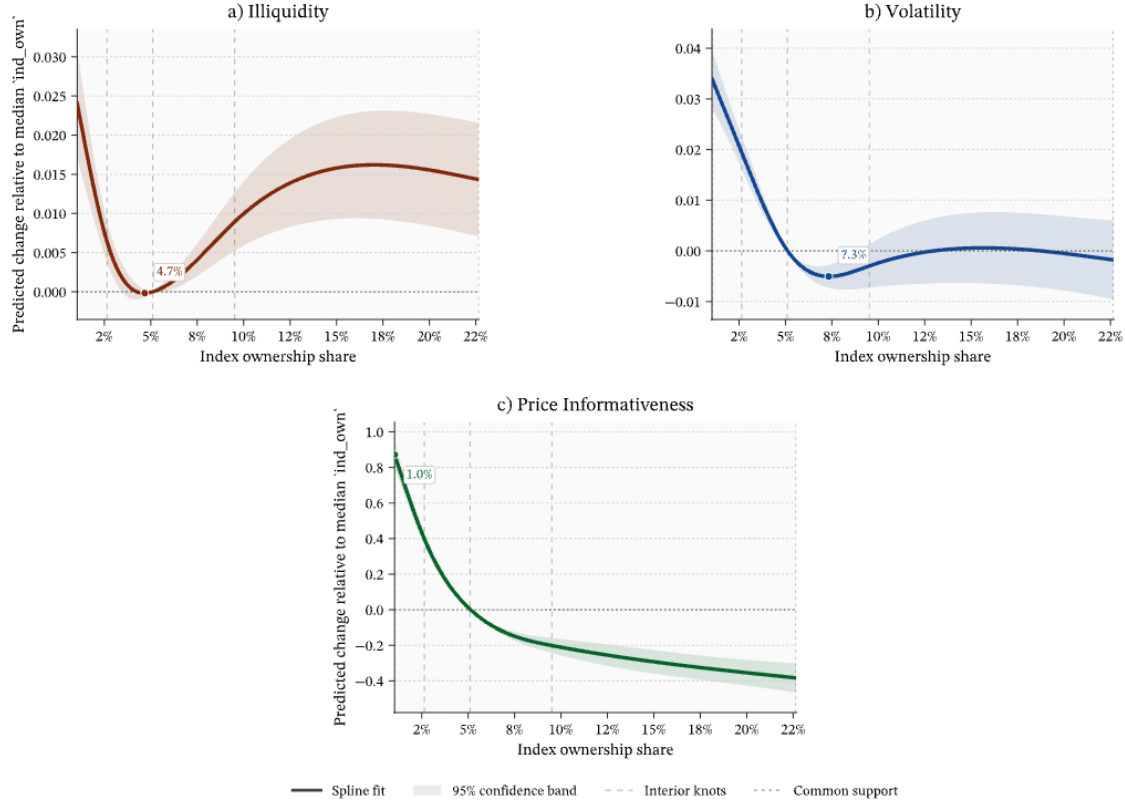


Figure 1.7: Reduced-Form Spline Estimates

Notes: This figure plots panel fixed-effects spline estimates of the relationship between index ownership and market quality. Curves are centered at the median index ownership, shaded bands denote point-wise 95% confidence intervals, and vertical dashed lines mark common support and interior knot locations. Labeled points indicate the outcome-specific optimum implied by the spline.

Figure 1.7 summarizes the channel-by-channel fitted curves, showing a clear non-linear relationship between index ownership and market quality. For Amihud illiquidity and volatility, the fitted curves improve as ownership moves out of the lower tail and into the moderate range. This is consistent with the interpretation that passive ownership can reduce trading frictions and stabilize prices at intermediate ownership levels. Price informativeness is weaker and less precisely pinned down, suggesting that the information channel is weaker than the liquidity and volatility channels in this specification. Taken together, the results imply that the effect of indexing is not

constant across the ownership distribution, but instead concentrated in the interior region where market functioning appears to be strongest.

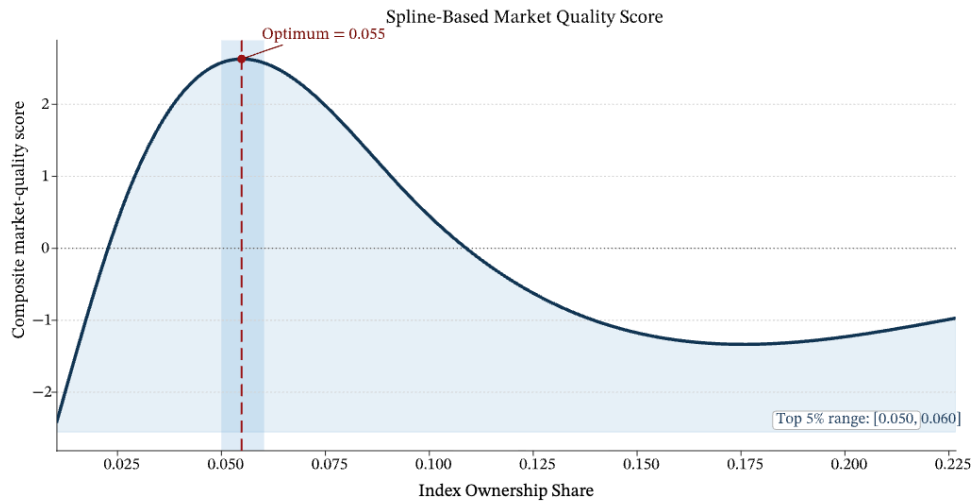


Figure 1.8: Reduced-Form Spline Market-Quality Score

Notes: This figure plots the composite market-quality score implied by the spline specification. Higher values indicate a more favorable combination of lower illiquidity, lower volatility, and higher price informativeness. The shaded region marks the top 5% of score values, and the vertical marker identifies the implied optimal range of index ownership.

Figure 1.8 traces the composite score across the ownership grid. In the stabilized reduced-form spline, the score peaks at approximately 0.0548, and the top 5% interval lies between 0.0500 and 0.0603. That narrow band indicates that the best market-quality outcomes are concentrated in a relatively tight interior region rather than spread broadly across moderate ownership levels. Economically, this is useful because it translates the separate channel responses into one summary measure of where the balance of liquidity, volatility, and informativeness is most favorable.

The score peak is also informative about how the channels interact. Liquidity and volatility improve most clearly in the moderate range, whereas price informativeness does not contribute as strongly to a sharp interior optimum. As a result, the composite

score should be interpreted as an economically weighted compromise across market-quality dimensions rather than as a universal threshold for indexing ownership. This is the main advantage of the score construction: it turns several related outcomes into a single market-quality summary while preserving the fact that the underlying channels do not move in identical ways.

Relative to the quadratic benchmark, the spline specification is valuable because it relaxes the assumption of one smooth global curvature and allows the data to reveal where the relationship bends most sharply. The estimated optimum moves only slightly upward, from about 0.045 in the quadratic exercise to about 0.055 in the stabilized spline, but the substantive conclusion does not change. In both cases, the evidence points to an interior range of ownership in which market quality is strongest. That stability is important because it indicates that the basic result is not driven by a restrictive polynomial form, even though the spline reveals more detail about how the contribution of indexing differs across the low, middle, and high parts of the support.

The spline comparison also clarifies how to read the shift in the estimated optimum. A quadratic fit imposes a single curvature over the full range, so it can summarize the average direction of the effect but not the local economic mechanism. The spline, by contrast, is better suited to identifying whether moderate indexing improves market quality because of a broad interior plateau or because of a sharper turning region. In that sense, the spline result strengthens the reduced-form argument by showing that the interior optimum persists when the functional-form restriction is relaxed, which is exactly the kind of robustness that matters in empirical finance.

1.5.5 SPLINE IV ANALYSIS

The spline analysis is also extended to an instrumental-variables setting using the global Russell 2000 cutoff design.

The spline IV specification extends the baseline IV framework by allowing the effect of index ownership to vary across the ownership distribution rather than imposing a single linear effect. The first stage uses the Russell 2000 cutoff-based instrument to isolate exogenous variation in index ownership, while the second stage uses the fitted ownership terms to estimate market-quality effects; the positive-part functions allow the slope to change at the prespecified knots. The two equations are specified as follows:

First Stage:

$$\begin{aligned}
IO_{iq} = & \pi_0 + \pi_1 Z_{iq} + \pi_2 (Z_{iq} - \kappa_1)^+ + \pi_3 (Z_{iq} - \kappa_2)^+ + \pi_4 (Z_{iq} - \kappa_3)^+ \\
& + \gamma X'_{iq} + \alpha_i + \lambda_q + \nu_{it}
\end{aligned} \tag{1.10}$$

Second Stage:

$$\begin{aligned}
Y_{iq} = & \beta_1 \widehat{IO}_{iq} + \beta_2 (\widehat{IO}_{iq} - \kappa_1)^+ + \beta_3 (\widehat{IO}_{iq} - \kappa_2)^+ + \beta_4 (\widehat{IO}_{iq} - \kappa_3)^+ \\
& + \delta X'_{iq} + \alpha_i + \lambda_q + \varepsilon_{iq}
\end{aligned} \tag{1.11}$$

In this extension, the endogenous spline basis in index ownership is instrumented with interactions of the cutoff indicator and running variable powers up to the fourth order, together with the standard firm-size, dollar-volume, and running-variable controls. The same spline knots are retained, and firm and date fixed effects are absorbed through two-way demeaning. This design is important because it allows the flexible spline shape to be estimated in a causal framework rather than only as a reduced-form correlation.

Table 1.6 reports the first-stage diagnostics for the endogenous spline basis. The first-stage diagnostics are strong across all three market-quality specifications, with minimum F-statistics remaining around 15.7 to 15.9 even after the endogenous spline basis is expanded into multiple terms. That is an important result because spline IV

Table 1.6: First-Stage F Statistics for Spline IV at Global Russell 2000 Cutoff

	Amihud illiquidity	Volatility	Price informativeness
Spline basis 1	53.9018 (p=<1e-16)	53.8301 (p=<1e-16)	53.9410 (p=<1e-16)
Spline basis 2	18.5412 (p=<1e-16)	18.4897 (p=<1e-16)	18.4978 (p=<1e-16)
Spline basis 3	15.7369 (p=1.55e-15)	15.9077 (p=9.99e-16)	15.6617 (p=1.89e-15)
Spline basis 4	56.1608 (p=<1e-16)	56.2035 (p=<1e-16)	56.0468 (p=<1e-16)
Observations	235544	234897	234281
Firms	8627	8605	8614
Dates	99	99	99
Cutoff	2000	2000	2000
Poly degree	4	4	4
Min first-stage F	15.7369	15.9077	15.6617
Controls	Yes	Yes	Yes
Fixed effects	firm + date	firm + date	firm + date
SE type	firm clustered	firm clustered	firm clustered

Notes: This table reports first-stage relevance diagnostics for the spline IV specification. Entries are first-stage F statistics for each endogenous spline basis term, with p-values shown underneath. The model uses firm and date fixed effects and firm-clustered standard errors. * p<0.10, ** p<0.05, *** p<0.01.

The spline basis is a natural cubic spline in ‘*ind_own*’ with four endogenous terms and excluded instruments formed by cutoff interactions with running-variable powers.

designs are typically harder to identify than simple linear specifications. The global Russell 2000 IV explains the ownership basis with enough power to support credible second-stage inference. In practical terms, this means the causal extension is not being pushed forward by a weak instrument or by a single favorable outcome; the relevance condition holds consistently across the market-quality measures. The strong first stage therefore gives the spline IV model a rare combination of flexibility and identification strength, making it a materially more convincing causal check than the local nonlinear IV specifications discussed earlier in the paper.

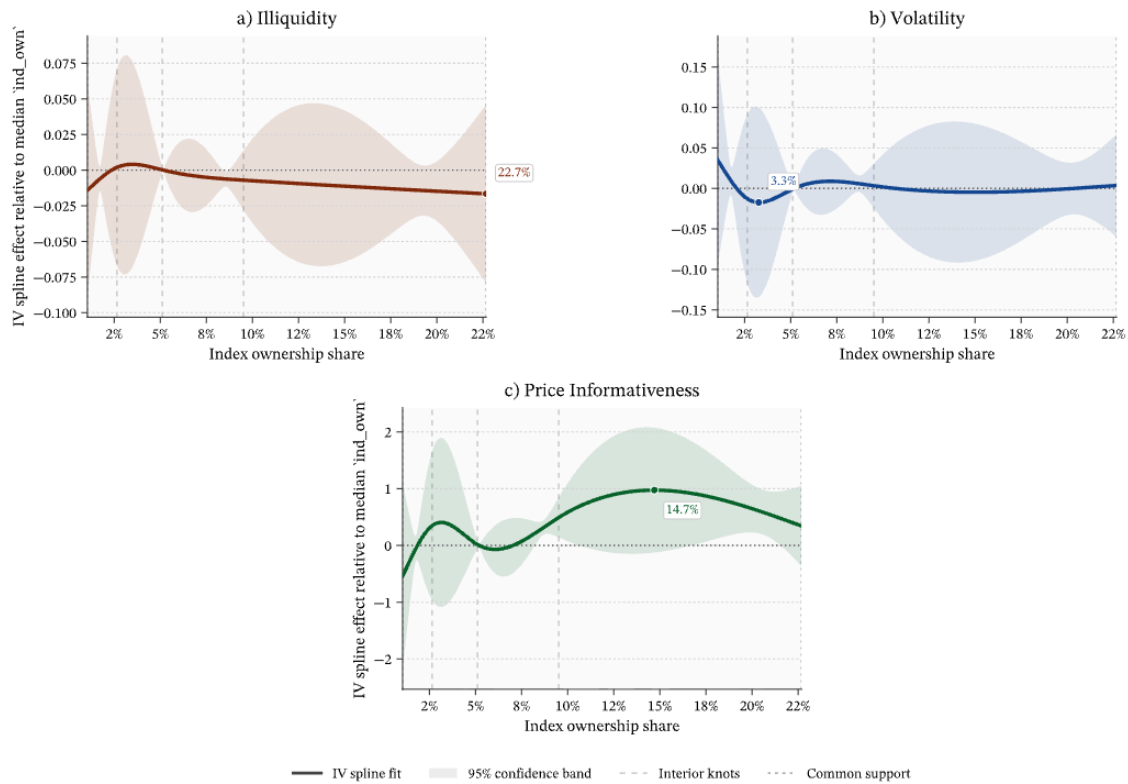


Figure 1.9: Spline IV Estimates

Notes: This figure shows IV spline estimates of the nonlinear effect of index ownership on each market-quality outcome under the global Russell cutoff 2000 design. Curves are centered at the median ind_own, shaded bands denote pointwise 95% confidence intervals, and vertical dashed lines mark spline knot locations. The figure highlights heterogeneous causal nonlinearities across liquidity, volatility, and price informativeness.

Figure 1.9 presents the spline IV estimates across the three market-quality measures. The fitted relationships are relatively flat for liquidity and volatility, whereas price informativeness exhibits more visible curvature, although the wide confidence bands indicate substantial estimation uncertainty. The fitted line it bends in a way that is consistent with an interior tradeoff between the benefits and limits of passive ownership. At lower ownership levels, additional indexing appears to improve market functioning. This is consistent with the idea that a larger passive presence can reduce trading friction, deepen liquidity provision, and dampen short-run price pressure. As indexing increases, the curve then flattens and eventually changes slope, which is the standard sign of diminishing marginal gains from additional passive ownership.

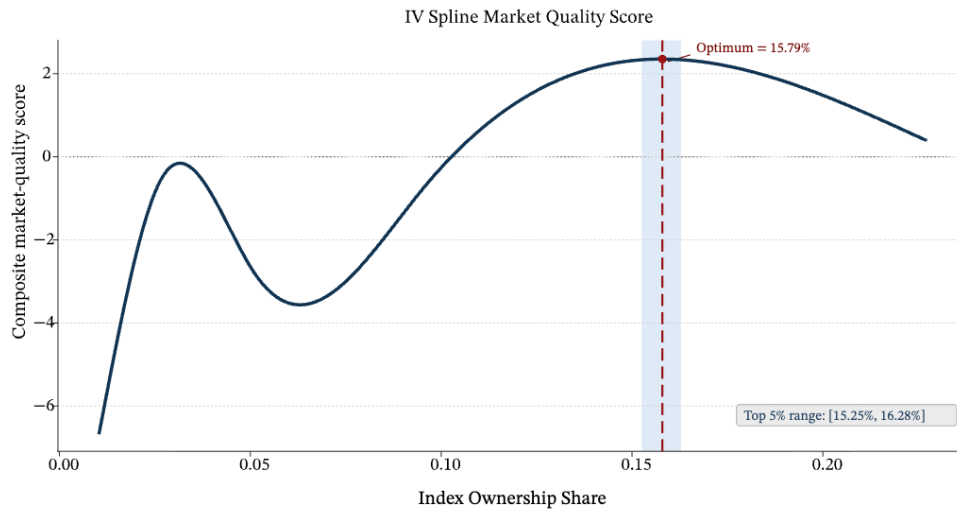


Figure 1.10: Spline IV Market-Quality Score

Notes: This figure reports the composite market-quality score based on the IV spline estimates under the global cutoff 2000 design. The peak of the curve identifies the causal ownership level associated with the strongest overall market-quality outcome, while the shaded band indicates the top 5% range of score values.

Figure 1.10 summarizes those fitted relationships in a single market-quality score. The composite optimum lies at approximately 0.1579, with the top 5% interval between 0.1525 and 0.1628. Visually, this confirms that the best causal market-quality

outcome is concentrated in a relatively narrow interior range rather than being spread across the ownership distribution. Another interesting takeaway is that the most favorable ownership region farther to the right than in the reduced-form benchmark. The key implication is that once endogeneity is removed, the strongest joint market-quality outcome appears at a higher ownership level. In other words, the market needs a larger passive presence before the combined liquidity and volatility gains dominate the tradeoff.

Taken together, the reduced-form spline and the causal spline IV extension support the same broad conclusion: indexing ownership is associated with market quality in a nonlinear way, and the best outcomes occur in an interior region of the ownership distribution. The reduced-form model places that region around 5% to 6% ownership, while the IV extension shifts the composite causal optimum upward toward roughly 15% to 16% ownership. The difference between the two estimates should not be read as a contradiction. Rather, it shows that the shape of the relationship is sensitive to endogeneity correction and the spline IV extension is valuable.

The main value of the spline IV extension is that it strengthens both identification and robustness. On the identification side, the preferred Russell 2000 design delivers strong first-stage relevance for the entire spline basis, so the second stage is strong and meaningful. On the robustness side, the spline relaxes the quadratic restriction and shows that the interior optimum survives when the functional form is allowed to vary more flexibly across the support. This combination matters because it rules out two common concerns at once: that the result is merely a polynomial artifact, and that the causal pattern is unstable once the ownership relation is estimated more carefully.

More broadly, the spline analysis indicates that the relationship between indexing and market quality is nonlinear and that the economically relevant ownership range is

interior rather than extreme. The spline analysis complements the quadratic analysis by confirming interior nonlinearity under a more flexible specification and by showing that the causal shape remains nonlinear once the design is extended to the Russell 2000 IV framework. This section strengthens the discussion as it adds credibility to both functional-form robustness and identification.

1.6 CONCLUSION

This dissertation examines how index ownership affects market quality across liquidity, volatility, and price informativeness using a range of econometric approaches. The baseline two-way fixed-effects regressions provide a descriptive benchmark: higher index ownership is associated with higher Amihud illiquidity ($\beta = 0.0613$), lower realized volatility ($\beta = -0.0560$), and lower values of the synchronicity-based informativeness proxy ($\beta = -3.3018$). These mixed signs indicate that passive ownership operates through multiple channels rather than a single monotonic mechanism. Accordingly, the baseline results motivate identification and functional-form analysis to distinguish correlation from causal effects and to test for nonlinear responses.

The instrumental-variables evidence uses Russell 1000/2000 cutoffs to isolate exogenous variation in ownership. At the Russell 1000 cutoff, index ownership is associated with a statistically precise reduction in volatility, while effects on illiquidity and informativeness are directionally consistent but imprecise, implying that the volatility channel is the most robust causal margin for large-cap firms. At the Russell 2000 cutoff, the ownership shock is larger and yields a sizable liquidity improvement ($\beta = -0.2849$) alongside a pronounced decline in the informativeness proxy ($\beta = -12.0868$), with volatility effects weaker. This pattern suggests that for smaller firms, passive inflows can deepen trading while simultaneously reducing firm-specific

information in prices. Taken together, the IV results indicate that causal effects vary across cutoffs and firm environments, and that liquidity gains are most evident for smaller firms experiencing larger passive inflows.

Nonlinear analysis refines the interpretation of these mixed linear results. Quadratic fixed-effects models reveal U-shaped patterns for volatility and price informativeness, while illiquidity exhibits weaker curvature; a composite market-quality score peaks at index ownership around 0.0379 with a high-quality range of roughly 3.3%~4.3%. IV-based nonlinear estimates confirm a robust U-shape for volatility at the Russell 2000 cutoff, with a turning point near 0.0809, while curvature in other channels is less stable under causal identification.

The spline analysis provides the strongest evidence that indexing ownership affects market quality in a nonlinear way. In the reduced-form spline specification, the composite market-quality optimum lies in a moderate ownership range of roughly 5% to 6% with the highest score concentrated in a narrow interval around 0.0500 to 0.0603. The spline IV extension strengthens the interpretation the causal interpretation: under the global Russell 2000 identification design, the nonlinear pattern remains robust after addressing endogeneity, preserving an interior optimum while shifting the favorable ownership range upward to approximately 15% to 16%, and maintaining a strong first stage across the full spline basis. Importantly, these results show that the interior optimum is not a quadratic artifact but a stable feature of the data that survives causal identification. Across both quadratic and spline specifications, a consistent conclusion emerges: the most favorable market outcomes occur in an interior ownership range, with volatility providing the clearest and most robust curvature and liquidity improving most clearly at moderate ownership levels.

This analysis has several limitations that qualify the interpretation of the results. First, index ownership is measured from index mutual funds and ETFs, so the analysis

may not capture passive ownership held through other investment vehicles. Second, the quarterly frequency may smooth short-run liquidity dynamics that occur within quarters. Third, while the Russell cutoff provides credible identification, the exclusion restriction is strongest near the cutoff and the nonlinear IV design remains sensitive to weak-instrument concerns for squared ownership. Fourth, the informativeness proxy is based on return synchronicity, which captures comovement but may not map one-to-one to firm-specific information production. Finally, the sample covers U.S. equities from 1993–2023, so external validity to other markets, asset classes, or regulatory regimes is uncertain.

Despite these limitations, the combined baseline, IV, and nonlinear evidence provides a coherent picture of how passive ownership relates to market quality and identifies the ranges where the trade-offs are most pronounced. Overall, the evidence points to a trade-off between liquidity provision and information production in the presence of passive ownership. Baseline correlations and causal estimates suggest that indexing can lower volatility and improve liquidity in some regimes, although the statistical evidence is mixed and relatively weak across specifications. The nonlinear results reconcile these tensions by showing that the net effect depends on the ownership level and firm environment, rather than a single average coefficient. This conclusion underscores that the market-quality effects of indexing are not uniform and that policy discussions should focus on marginal changes around prevailing ownership levels rather than on extremes.

CHAPTER 2

HETEROGENEOUS EFFECTS OF INDEX OWNERSHIP: EVIDENCE FROM FIRM SIZE AND BUSINESS CYCLE CONDITIONS

2.1 INTRODUCTION

The growth of index ownership has altered the investor base of U.S. equities and, in turn, raised a core question for market quality: when does passive ownership improve trading conditions and when does it weaken information production? The baseline regressions show mixed average effects across liquidity, volatility, and a synchronicity-based informativeness proxy, which indicates that a single average coefficient is unlikely to capture the full pattern. As a result, this chapter maps how ownership–market-quality correlations vary across environments. The objective is descriptive rather than structural, and the analysis is designed to provide additional evidence on where the trade-offs are strongest rather than to identify an optimal ownership level.

The heterogeneity analysis proceeds along three dimensions. First, the size-sorted design tests whether ownership effects differ across the firm-size distribution, where information frictions and trading depth vary systematically. Second, the business-cycle design evaluates whether ownership effects shift between expansions and recessions, when market-wide liquidity and risk conditions change sharply. Third, the combined size–cycle design examines joint heterogeneity to determine whether size gradients are

amplified or dampened in downturns. This progression provides a structured mapping of ownership effects across the cross-section and the time series.

A large literature highlights that the effects of index ownership on market quality are heterogeneous across firms, particularly by size. Large-cap firms in major indices often experience improved liquidity and narrower spreads, while small-cap firms may face weaker or adverse effects due to lower trading volumes and market depth (Ben-David, Franzoni, and Moussawi, 2018; Petajisto, 2013; Cremers and Petajisto, 2009). Firm size also shapes trading dynamics, with larger firms attracting more informed and institutional trading that improves price efficiency (Chordia, Roll, and Subrahmanyam, 2000; Hasbrouck, 2007). To capture this heterogeneity, scholars employ size-sorted regressions that estimate ownership effects within deciles or quintiles while controlling for firm and time factors (Brennan and Subrahmanyam, 1996; Foucault, Kadan, and Kandel, 2005; Acharya et al., 2013). This approach provides a detailed view of how indexing effects vary across the size distribution.

A related literature emphasizes state dependence over the business cycle. Market liquidity, volatility, and investor risk appetite shift between expansions and recessions, which can amplify or mitigate passive-ownership effects (Foucault, Sraer, and Thesmar, 2010; Ben-David, Franzoni, and Moussawi, 2018; Adrian, Boyarchenko, and Giannone, 2019). Expansions tend to absorb passive flows with limited disruption, whereas recessions or financial stress can exacerbate volatility and bid-ask spreads, especially for smaller or less liquid firms (Bekaert, Harvey, and Lundblad, 2007; Kelly, Pruitt, and Su, 2018). Market microstructure research also shows that liquidity provision and trading dynamics vary with macro conditions (Brunnermeier and Pedersen, 2009; Acharya et al., 2013). These findings underscore the need to model cyclical heterogeneity when evaluating index ownership.

This study contributes to the literature on index ownership and market quality in three ways. First, it extends prior work on firm-size heterogeneity by systematically analyzing how passive ownership affects liquidity, volatility, and price informativeness across size deciles. Second, it incorporates business-cycle dynamics to test whether ownership effects shift between expansions and recessions, highlighting state-dependent market responses. Third, and most importantly, it integrates these two dimensions by examining how the interplay between firm size and macro conditions shapes the impact of indexing on market quality. By jointly considering size and cyclical effects, the analysis provides a more nuanced answer to the question of how much indexing is good for the market and informs both academic debates and policy discussions.

The empirical strategy builds on the baseline two-way fixed-effects framework with controls for firm size and dollar trading volume. For size heterogeneity, the specification is estimated separately within market-capitalization deciles. For cycle heterogeneity, index ownership is interacted with a recession indicator to capture slope shifts between downturns and expansions. For joint heterogeneity, ownership effects are estimated within each size decile and cycle segment, producing a grid of coefficients summarized by heatmaps. All specifications retain firm and time fixed effects and two-way clustered standard errors.

The remainder of the chapter proceeds as follows. Section 2.1 presents size-sorted results, highlighting stronger informativeness effects among smaller firms and limited liquidity effects. Section 2.2 examines business-cycle heterogeneity, showing that liquidity and volatility vary across regimes while informativeness remains more stable. Section 2.3 combines size and cycle conditions and finds that liquidity and volatility effects are concentrated in particular size-cycle groups, while informativeness remains broadly negative. Section 2.4 concludes with limitations.

2.2 SIZE-SORTED ANALYSIS

2.2.1 SET-UP

Firm size is a core determinant of trading intensity and information environments. Smaller firms face higher information frictions, thinner liquidity, and more segmented investor bases, whereas larger firms trade more frequently and are covered by broader information intermediaries. Consequently, identical changes in passive ownership may trigger different responses in liquidity provision, volatility, and price informativeness across size groups.

Figure 2.1 presents a stylized fact that motivates the size-sorted analysis. Index ownership generally increases from the smallest to the largest market-capitalization deciles, although the pattern is not strictly monotonic among the largest firms. This cross-sectional variation motivates testing whether the relationship between index ownership and market quality differs systematically with firm size.

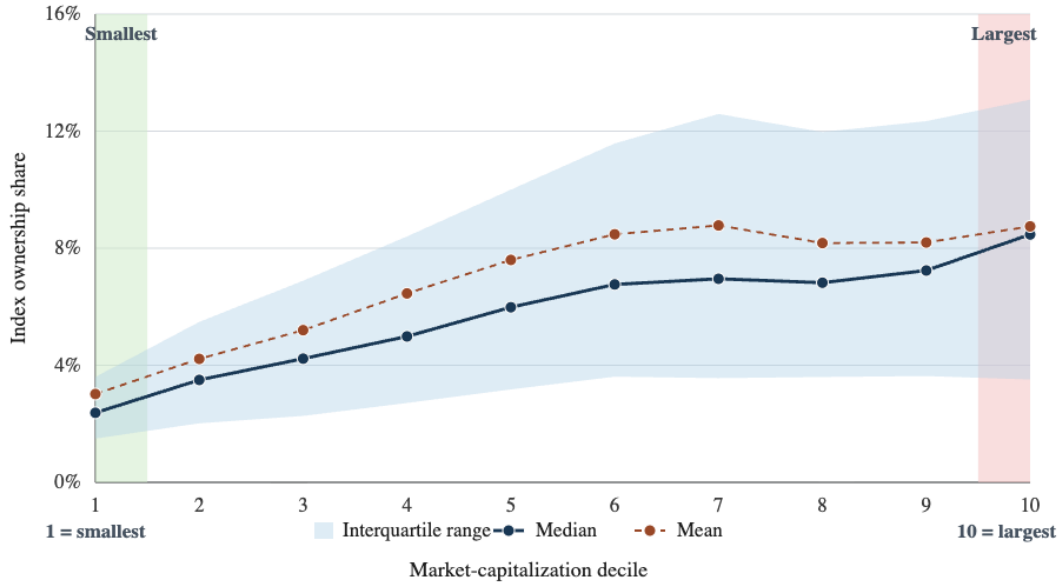


Figure 2.1: Size-Sorted Index Ownership

Notes: Index ownership by market-capitalization decile. The solid line reports the median and the dashed line reports the mean. The shaded area represents the interquartile range.

The previous chapter documents that index ownership is associated with multiple, and sometimes opposing, dimensions of market quality. Among the causal specifications, the spline IV analysis is preferred because it allows the ownership–market-quality relationship to vary flexibly across the observed ownership range. In the quadratic IV analysis, ownership effects differ across the Russell 1000 and Russell 2000 cutoffs: near the Russell 1000 cutoff (large-cap boundary), the estimates indicate a statistically precise decline in volatility, whereas near the Russell 2000 cutoff (small-cap boundary), they indicate improved liquidity alongside a pronounced decline in the informativeness proxy.¹ This evidence motivates a systematic heterogeneity analysis that conditions on firm size to identify more precisely where the impact of index ownership changes.

¹The corresponding quadratic IV results for the Russell 1000 and Russell 2000 cutoff specifications are reported in the Appendix under “Quadratic IV R1000/R2000–Full Results.”

This section evaluates heterogeneity in the relationship between index ownership and market quality across firm size. The heterogeneous analysis uses size-sorted regressions to map the ownership–market-quality relationship across the full distribution of market capitalization. It provides an average association, while the size-sorted design asks whether the same mechanisms operate for small and large firms with distinct information environments and trading frictions. The analysis sorts observations into market-capitalization deciles and estimates identical fixed-effects regressions within each decile. This approach isolates cross-sectional heterogeneity while holding the specification constant, so any variation in coefficients can be attributed to size-related differences.

The empirical design mirrors the baseline specification while allowing the coefficient on index ownership to vary by firm size.

$$Y_{i,q} = \alpha_i + \lambda_q + \beta_1 IndexOwn_{i,q} + \sum_{k=2}^{10} \delta_k (IndexOwn_{i,q} \times D_{k,i,q}) + \gamma X'_{i,q} + \epsilon_{i,q},$$

where

$$X'_{i,q} = [\ln(DollarVolume_{i,q})],$$

$$D_{k,i,q} = \mathbf{1}\{\text{firm } i \text{ belongs to decile } k \text{ in quarter } q\}.$$
(2.1)

The interaction term $(IndexOwn_{i,q} \times D_{k,i,q})$ allows the coefficient on index ownership to vary across firm-size deciles. Decile 1 is the omitted reference group, so β_1 represents the ownership slope for the smallest firms, while each δ_k measures the difference between decile k and decile 1. The nine interaction terms therefore generate ten decile-specific ownership slopes without creating additional groups or multiplying the sample.

The sample covers 1993Q1 to 2023Q4 and is partitioned into ten market-capitalization deciles, each containing approximately 23,900 observations. Sorting is performed at the observation level using market capitalization, ensuring that the within-decile regressions compare firms of similar size at each date.

Variable construction follows the baseline chapter. Amihud illiquidity is computed from daily returns and dollar volume and aggregated to the quarter; volatility is the quarterly standard deviation of daily returns; and price informativeness measure is the synchronicity-based price-informativeness proxy derived from a factor-model R^2 transformation. Index ownership uses quarter-end holdings of index mutual funds and ETFs relative to shares outstanding. The same data sources and filters apply, so size-based comparisons reflect heterogeneity rather than measurement changes.

The regressions use the ownership-covered subsample with consistent coverage for the three outcomes. Because the same specification is estimated within each decile, differences in coefficient patterns are not driven by model variation or uneven sample size. At the same time, these estimates remain correlational and should be interpreted as descriptive heterogeneity rather than causal effects.

2.2.2 SIZE EFFECTS

Figure 2.2 presents the size-sorted regression estimates for the effect of index ownership on the price-informativeness proxy. The size gradient is most pronounced for this outcome. Index ownership is negatively associated with the price-informativeness proxy in deciles 1 through 9, with large t-statistics in the small-cap tail (for example, $\beta = -13.60$ and $t = -10.47$ in decile 1, and $\beta = -8.21$ and $t = -12.47$ in decile 2). The magnitude attenuates steadily as size increases, and the coefficient becomes small and statistically insignificant in the largest decile ($\beta = -0.40$ and $t = -0.50$). This pattern indicates that the negative association between passive ownership and

the informativeness proxy is concentrated among smaller and mid-sized firms rather than among the largest firms.

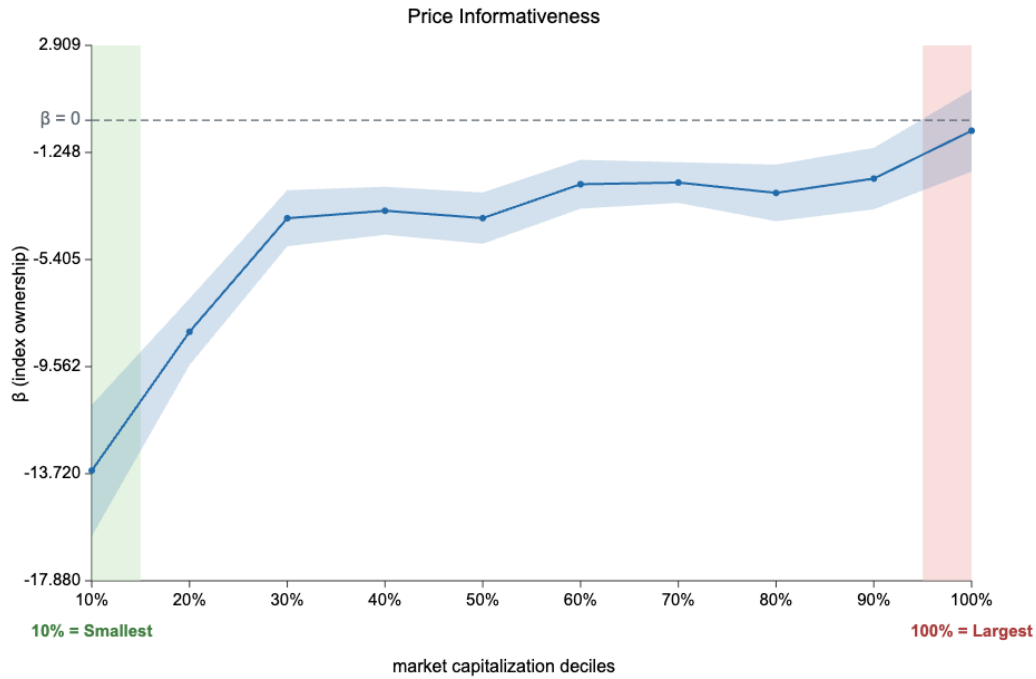


Figure 2.2: Size-Sorted Price Informativeness Estimates

Notes: Size-sorted coefficient estimates for index ownership on the price informativeness proxy with 95% confidence intervals.

The economic interpretation is consistent with the mechanism highlighted in the prior chapter: passive ownership can reduce incentives for firm-specific information production, and this effect appears strongest where baseline information frictions are high. Because small firms rely more heavily on marginal information producers, a shift toward passive ownership may disproportionately reduce the incorporation of firm-specific signals into prices. By contrast, for large firms with deep analyst coverage and abundant public information, the same ownership changes appear less consequential for the informativeness proxy. This size pattern therefore reconciles the baseline negative coefficient with the IV evidence at the Russell 2000 cutoff, where the

informativeness decline is most pronounced. At the same time, the findings underscore that the informativeness channel is not uniform across the cross-section.

Figure 2.3 presents the size-sorted regression estimates for the effect of index ownership on Amihud illiquidity. Liquidity effects, measured by the Amihud illiquidity proxy, are comparatively weak across size groups. Only the third decile exhibits a statistically significant estimate ($\beta = 0.019$ and $t = 2.28$), while the remaining deciles yield economically small and statistically indistinguishable coefficients. The coefficient range from -0.068 to 0.019 suggests that any size-related variation in the illiquidity response is modest relative to the variation observed in the informativeness proxy. This lack of a clear monotone pattern contrasts with the IV evidence near the Russell 2000 cutoff, where liquidity improvements were sizable, and indicates that such effects may be localized to specific ownership shocks rather than broad size categories.

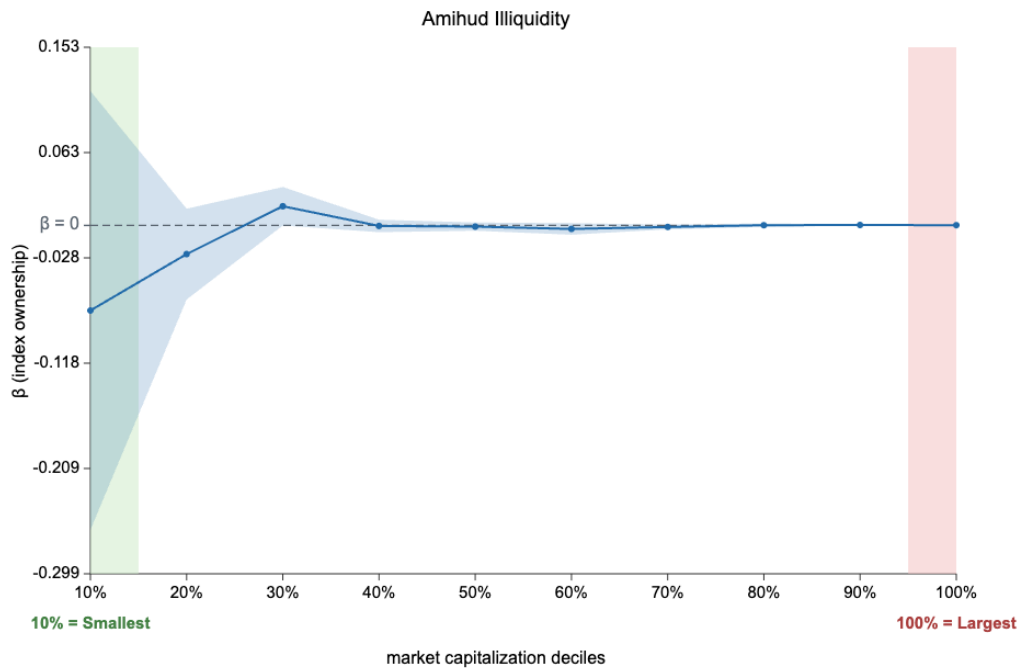


Figure 2.3: Size-Sorted Amihud Illiquidity Estimates

Notes: Size-sorted coefficient estimates for index ownership on Amihud illiquidity with 95% confidence intervals.

One interpretation is that the liquidity response associated with passive ownership depends more on the size and timing of ownership changes and the broader market environment than on firm size itself. If passive ownership changes are concentrated around index reconstitutions or among particular investor segments, the liquidity response may be shaped by the specific ownership change rather than by firm size alone. Alternatively, the Amihud proxy may be less sensitive to small changes in ownership within a size group once firm and date fixed effects are included. In either case, the size-sorted design suggests that liquidity is the least systematic channel in the cross-section relative to informativeness and volatility. This conclusion highlights the value of complementing size-sorted regressions with event-based identification for liquidity effects.

Figure 2.4 presents the size-sorted regression estimates for the effect of index ownership on idiosyncratic volatility. The estimates are negative across much of the lower and middle size distribution, but statistical significance is concentrated in deciles 4–6. Deciles 4 through 6 show negative and significant coefficients (β between -0.139 and -0.106, with t -statistics around -2.1 to -2.3), indicating lower idiosyncratic volatility as index ownership rises for mid-cap firms. The smallest decile has a similar sign but only marginal significance ($t = -1.93$), while the largest deciles show small and statistically insignificant positive coefficients. This pattern suggests a localized mid-cap association rather than a monotonic relationship between index ownership and volatility.

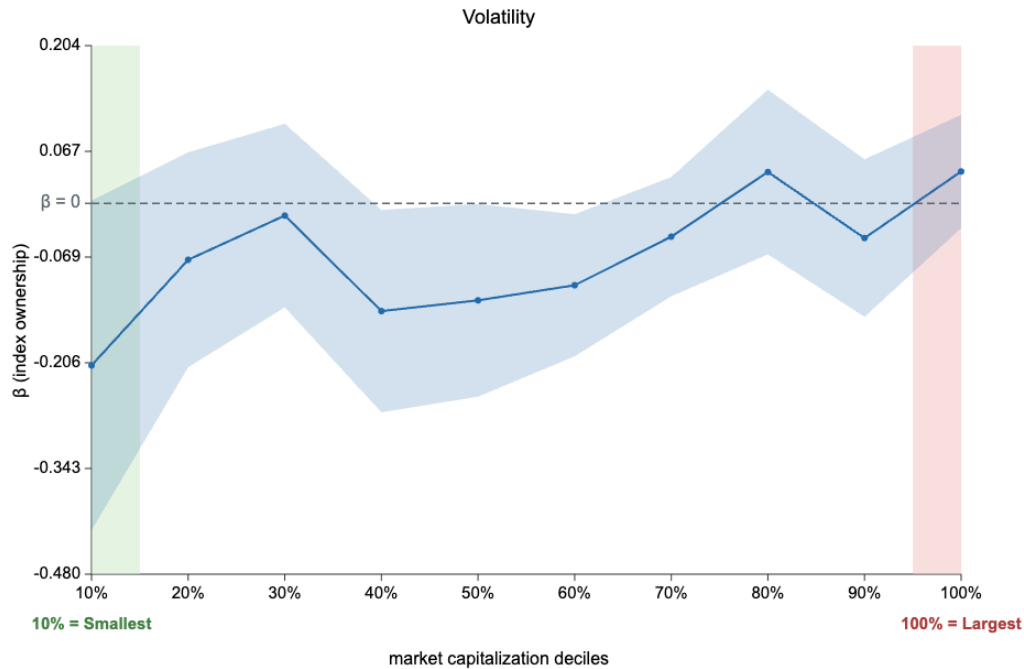


Figure 2.4: Size-Sorted Volatility Estimates

Notes: Size-sorted coefficient estimates for index ownership on idiosyncratic volatility with 95% confidence intervals.

The mid-cap emphasis also aligns with the prior chapter’s IV evidence at the Russell 1000 cutoff, which identified volatility reductions for large-cap firms but weaker effects at other cutoffs. In the size-sorted results, the volatility effect is not monotone with size, implying that the ownership–volatility relationship depends on a balance between liquidity depth and information sensitivity. This interpretation is consistent with a view in which passive ownership dampens idiosyncratic price movements by increasing common ownership and stabilizing flows, but only within a particular range of firm size. Outside that range, either trading is too thin for flow-based stabilization or information production is too robust for ownership shifts to materially alter volatility. The nonlinearity across size reinforces the broader conclusion that ownership effects are context dependent.

The size-sorted evidence indicates that the informativeness channel is the most consistently heterogeneous dimension of market quality. It suggests that size interacts more strongly with information channels than with trading-cost channels. Small firms typically face thinner analyst coverage and higher idiosyncratic information frictions, so index ownership may affect the price-information proxy more strongly in those segments. Liquidity in large firms is already high and may be less sensitive to changes in index-fund ownership, which is consistent with flat illiquidity coefficients. The volatility pattern aligns with diminishing marginal effects, where incremental index ownership may stabilize returns when baseline volatility is high but becomes negligible for large firms. In economic terms, the most meaningful size-related pattern is the concentration of informativeness effects among smaller firms, while volatility stabilization is narrower and liquidity effects are not systematic.

While the size-sorted analysis is informative, it potentially misses another important factor: cyclical effects. The influence of index ownership on market conditions may vary with broader macroeconomic and financial environments, and market quality can shift sharply across phases of the business cycle. Liquidity and volatility are particularly cyclical, which implies that the same ownership change may have different implications in expansions than in recessions. As a result, a business-cycle-based heterogeneous analysis is conducted to examine how the ownership–market-quality relationship differs across expansion and recession regimes.

2.3 BUSINESS CYCLE ANALYSIS

2.3.1 SET-UP

Business-cycle conditions are a central determinant of market quality, and they provide a natural dimension of heterogeneity for the effects of index ownership. Liq-

uidity, volatility, and the informational content of prices move with macroeconomic conditions, and these shifts can alter how passive ownership translates into trading outcomes. The time-series summaries from the data description section in previous chapter indicate that market-quality proxies and index ownership display pronounced variation across the sample, which motivates an explicit interaction between ownership and recession states. Accordingly, this section asks whether the association between index ownership and market quality differs in recession versus expansion regimes. By conditioning on business-cycle states, the analysis complements the size-sorted results and clarifies whether ownership effects are amplified or muted when market conditions deteriorate.

The recession indicator follows NBER dates and covers three episodes. The recession periods are 2001Q1–2001Q4, 2007Q4–2009Q2, and 2019Q4–2020Q2. These episodes coincide with periods of heightened macroeconomic stress and substantial dislocations in trading activity. In the time-series summaries, the ownership measure and market-quality proxies display clear movements over time, reinforcing the need to allow ownership effects to differ across these regimes. This background motivates a panel specification that interacts index ownership with a recession dummy while retaining firm and time fixed effects. The goal is to test whether the ownership–market-quality relationship shifts in economically meaningful ways during downturns.

The business-cycle heterogeneity specification follows the baseline two-way fixed-effects design but adds an interaction between index ownership and a recession indicator.

$$Y_{i,q} = \alpha_i + \lambda_q + \beta IndexOwn_{i,q} + \delta (IndexOwn_{i,q} \times R_q) + \gamma X'_{i,q} + \varepsilon_{i,q},$$

where

$$X'_{i,q} = [c_firm_size_{i,q}, c_dollar_vol_{i,q}],$$

$$R_q = \mathbf{1}\{\text{quarter } q \text{ is a recession quarter}\}.$$
(2.2)

Because $R_q = 0$ during expansions, β represents the ownership slope in expansions. The interaction coefficient δ measures the incremental change in this slope during recessions. Accordingly, the recession-state slope is $\beta + \delta$. The recession indicator itself is absorbed by the time fixed effects, so only its interaction with index ownership is identified. The control vector includes the firm-size and dollar-volume controls used in the baseline specification.

The regression is estimated separately for Amihud illiquidity, idiosyncratic volatility, and the synchronicity-based price informativeness proxy. The sample contains approximately 239,758–241,199 firm-quarter observations with about 8,855–8,877 firms and 100 time periods, depending on the outcome. Because the model includes time fixed effects, the main recession dummy is absorbed for identification of the interaction, and the coefficients should be interpreted as within-firm slope shifts between recessions and expansions. The results section reports the expansion-state slopes and the incremental recession adjustments.

2.3.2 CYCLIC EFFECTS

To visualize how the ownership–market-quality association evolves over time, the analysis also estimates rolling 16-quarter panel fixed-effects regressions. The rolling specification provides a descriptive summary of local coefficient variation and allows the estimates to be compared with the NBER recession dates shown by the shaded

regions. The series does not begin in the earliest expansion window, 1993Q1–2000Q4, because a sufficiently populated 16-quarter estimation window is required after applying the sample filters. These rolling estimates are descriptive and should not be interpreted as replacements for the full-sample interaction results. The main inference comes from the full-sample panel fixed-effects regressions with recession interactions reported in Table B.2, where β measures the expansion slope and δ captures the incremental recession adjustment.

The liquidity channel exhibits the strongest business-cycle heterogeneity. In expansions, higher index ownership is associated with higher Amihud illiquidity ($\beta = 0.080$ and $t = 3.37$), indicating weaker liquidity when passive ownership rises. During recessions, the interaction term is negative and statistically significant ($\delta = -0.2142$, $t = -2.47$), implying that the ownership–illiquidity slope is substantially lower in downturns. Taken together, the recession-state slope equals the sum of β and δ , which implies a materially different relationship between ownership and liquidity in recessions than in expansions. Figure 2.5 plots the rolling 16-quarter panel fixed-effects estimates for index ownership and Amihud illiquidity, with 95% confidence intervals. The shaded regions indicate NBER recession periods.

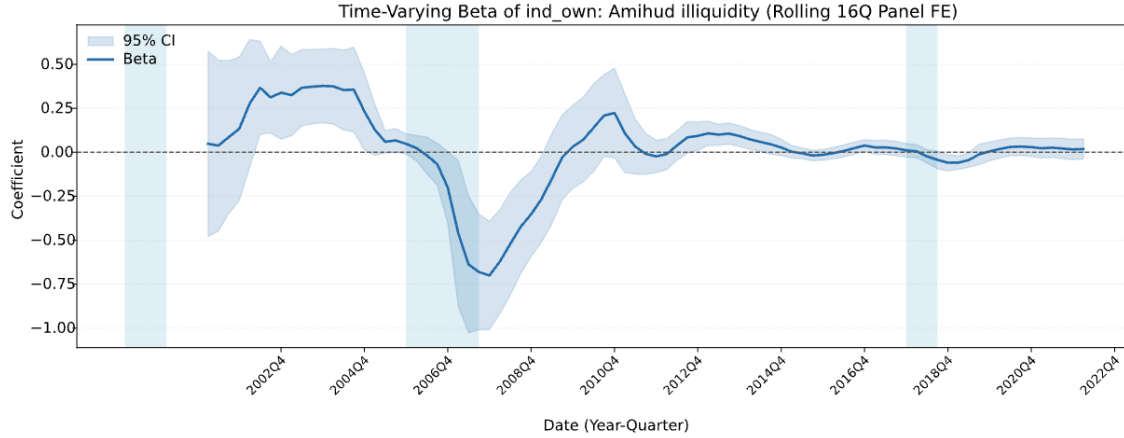


Figure 2.5: Rolling Panel FE Estimates for Amihud Illiquidity

Notes: Rolling 16-quarter panel fixed-effects estimates for index ownership on Amihud illiquidity with 95% confidence intervals. Shaded regions indicate NBER recession periods.

A plausible interpretation is that passive ownership interacts with macro liquidity conditions. When trading is strained in recessions, index ownership may provide a stabilizing base of demand or facilitate rebalancing activity that offsets adverse liquidity pressures. By contrast, in expansions the same ownership changes may coincide with a gradual shift away from active trading that increases illiquidity on the margin. This recession-specific attenuation is consistent with the notion that the liquidity channel is state dependent and should not be summarized by a single average coefficient. The next sections examine whether similar state dependence appears in volatility and informativeness.

For idiosyncratic volatility, the expansion-state coefficient is negative and statistically significant ($\beta = -0.0681$ and $t = -2.95$), indicating lower volatility as index ownership rises in normal times. The recession interaction is positive but statistically imprecise ($\delta = 0.1390$, $t = 1.34$), implying that the volatility-reduction effect weakens in downturns. While the interaction is not significant, the sign suggests that the stabilizing effect of passive ownership is less reliable when macro conditions deteriorate.

This pattern aligns with the broader thesis that ownership effects depend on the surrounding market environment rather than being constant over time. Figure 2.6 plots the corresponding rolling estimates for idiosyncratic volatility, with 95% confidence intervals and recession periods shown by the shaded regions.

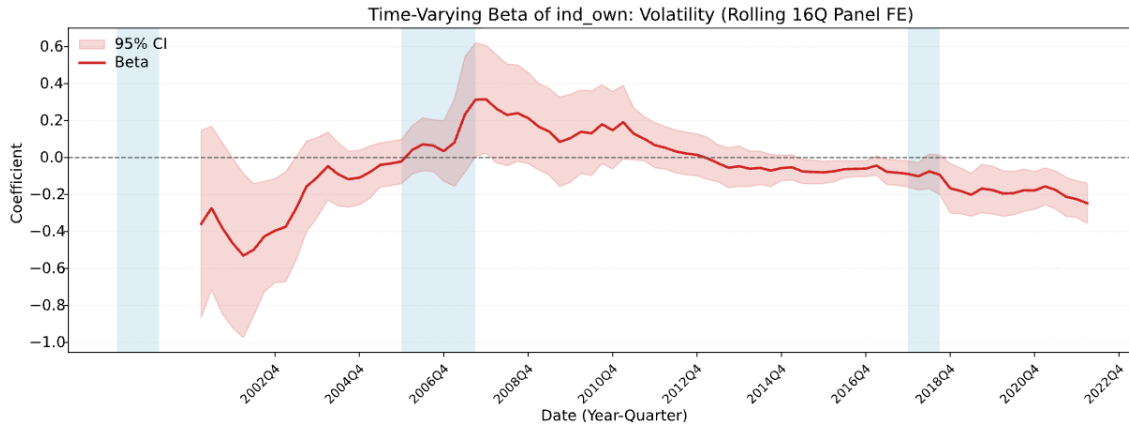


Figure 2.6: Rolling Panel FE Estimates for Idiosyncratic Volatility

Notes: Rolling 16-quarter panel fixed-effects estimates for index ownership on idiosyncratic volatility with 95% confidence intervals. Shaded regions indicate NBER recession periods.

The attenuation of volatility effects in recessions is economically intuitive. Periods of stress often involve large common shocks, correlated trading, and elevated risk premia that dominate firm-specific ownership changes. Under such conditions, the ability of passive ownership to damp idiosyncratic volatility may be limited by broader market forces. Conversely, in expansions when common shocks are smaller, ownership shifts may more effectively reduce firm-level volatility. This explanation highlights why the volatility channel appears more state dependent than the baseline average suggests.

The price informativeness proxy shows a negative association with index ownership in expansions ($\beta = -3.2530$, $t = -13.55$), consistent with the baseline evidence that higher passive ownership coincides with less informative prices. The recession interaction term is also negative but statistically insignificant ($\delta = -0.5590$, $t = -0.70$),

implying that the informativeness decline does not materially differ between recessions and expansions. This result suggests that the information-production channel operates persistently across macro regimes, rather than being concentrated in downturns. Figure 2.7 presents the rolling estimates for the synchronicity-based price informativeness proxy, again with 95% confidence intervals and shaded NBER recession periods.

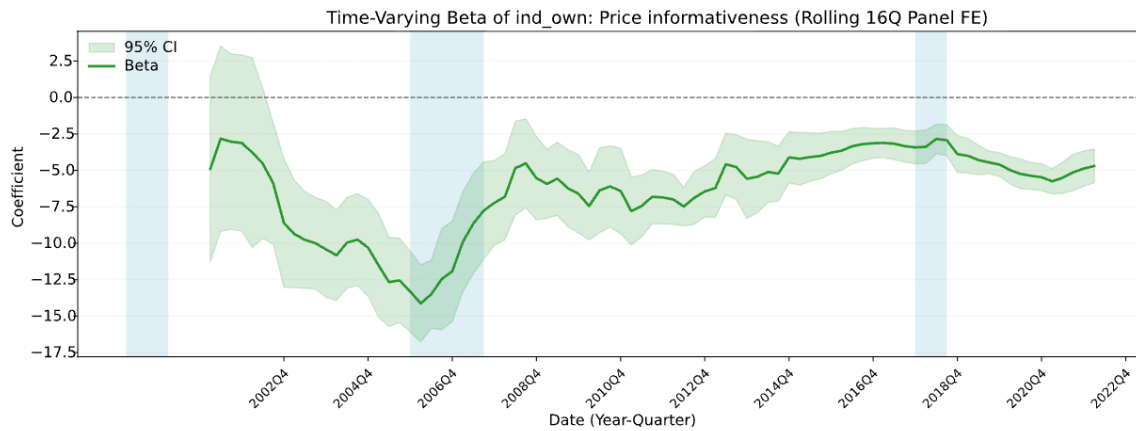


Figure 2.7: Rolling Panel FE Estimates for Price Informativeness

Notes: Rolling 16-quarter panel fixed-effects estimates for index ownership on the synchronicity-based price informativeness proxy with 95% confidence intervals. Shaded regions indicate NBER recession periods.

One interpretation is that information production responds to ownership structure more than to cyclical conditions. If passive ownership reduces incentives for firm-specific information acquisition in a persistent way, then recession conditions may not further amplify the effect. Alternatively, measurement limitations of the synchronicity proxy may attenuate state dependence, since macro shocks already raise comovement across firms. These considerations indicate that the informativeness channel is robust but relatively insensitive to cyclical regimes, in contrast to the liquidity channel. This distinction is important for interpreting the overall market-quality trade-off over time.

The business-cycle heterogeneity results provide the strongest evidence of regime variation in the liquidity channel, whereas the informativeness channel appears relatively stable across regimes. In expansions, higher index ownership coincides with higher illiquidity and lower volatility, whereas in recessions the ownership–illiquidity slope is substantially lower and the volatility effect weakens. The informativeness proxy remains negatively associated with ownership in both regimes, suggesting a persistent information-production trade-off. Economically, liquidity is the most state-dependent margin, whereas the persistent informativeness decline represents the most robust cross-regime trade-off; volatility stabilization is narrower because it weakens in recessions. This time-variation complements the size-sorted results and indicates that policy discussions should be conditioned on prevailing market states.

The business-cycle analysis reveals systematic variation in ownership effects across macro regimes, while the size-sorted analysis in the previous section reveals cross-sectional differences across firms. Taken together, these two dimensions of heterogeneity may provide a descriptive answer to the question: how much indexing is good for the market. This answer does not imply a structural optimum, but it motivates a heterogeneous analysis that conditions simultaneously on both firm size and business-cycle state. The next section would test whether the recession–expansion slope shifts are concentrated among small firms, mid-cap firms, or large firms, and whether size gradients are amplified or dampened in downturns.

2.4 SIZE AND BUSINESS-CYCLE INTERACTIONS

2.4.1 INTERACTION SET-UP

Size-sorted results indicate that the informativeness channel is strongest among small and mid-sized firms, while the business-cycle analysis shows that liquidity and volatil-

ity responses shift across expansions and recessions. This section therefore combines both dimensions and examines whether ownership effects differ jointly by firm size and by cycle regime. The exercise is descriptive rather than structural; its goal is to provide additional evidence on when indexing appears most associated with better or worse market conditions.

This joint heterogeneity design is motivated by the idea that the impact of passive ownership is not uniform across either firms or macro states. Smaller firms have higher information frictions and thinner trading, while recessions bring broad liquidity and risk shocks that can overwhelm firm-level dynamics. If the ownership–market–quality relationship is state dependent, then averaging across both dimensions may obscure the environments where the trade-offs are most pronounced. Thus, the combined analysis addresses a descriptive version of the question, “how much indexing is good for the market,” by identifying where the correlations are strongest and where they attenuate. The analysis does not claim an optimal structural level of indexing but instead maps ownership effects across size and cycle regimes.

The empirical strategy extends the baseline two-way fixed-effects regression by interacting index ownership with both firm-size deciles and business-cycle regimes. For each outcome, the analysis estimates separate slopes for each size decile within each cycle segment, using the same controls for firm size and dollar trading volume and retaining firm and time fixed effects. Business-cycle segments follow the NBER dates used earlier and include the recession intervals 2001Q1–2001Q4, 2007Q4–2009Q2, and 2019Q4–2020Q2, as well as surrounding expansion windows.

$$\begin{aligned}
Y_{iq} = & \sum_{d=1}^{10} \sum_{b \in \{0,1\}} \beta_{db} \left(\text{IndexOwn}_{iq} \times \mathbf{1}\{\text{SizeDecile}_i = d\} \times \mathbf{1}\{BC_t = b\} \right) \\
& + \delta_i + \gamma_q + \mathbf{X}'_{iq} \theta + \epsilon_{iq},
\end{aligned} \tag{2.3}$$

where

$\mathbf{1}\{\text{SizeDecile}_i = d\}$ = indicator for firm i belonging to size decile d ,

$\mathbf{1}\{BC_t = b\}$ = business-cycle regime (e.g., expansion: $b=0$, recession: $b=1$)

The resulting grid of coefficients can be summarized as a heatmap of t-statistics and by highlighting the most precisely estimated cells within each outcome. This approach emphasizes where ownership effects are most reliably estimated, while still acknowledging that many size-cycle cells have small samples, especially in short recession windows. Because the design remains descriptive, the coefficients should be interpreted as conditional correlations rather than causal effects. The next sections report the main patterns for liquidity, volatility, and price informativeness.

2.4.2 SIZE- AND CYCLE-DEPENDENT EFFECTS

Liquidity effects exhibit meaningful variation across both size and cycle regimes. The most precisely estimated liquidity cell occurs in the 2002Q1–2007Q3 expansion for size decile 7, where index ownership is associated with higher Amihud illiquidity ($\beta = 0.0348$, $t = 3.58$). In the same expansion window, several mid-to-large deciles also show positive and statistically significant estimates, whereas the largest decile displays a negative estimate. This pattern suggests that during mid-2000s expansions, higher ownership was correlated with weaker liquidity in a range of mid-sized firms but not uniformly across the distribution. Figure 2.8 visualizes the size-cycle liquidity pattern using the coefficient heatmap.

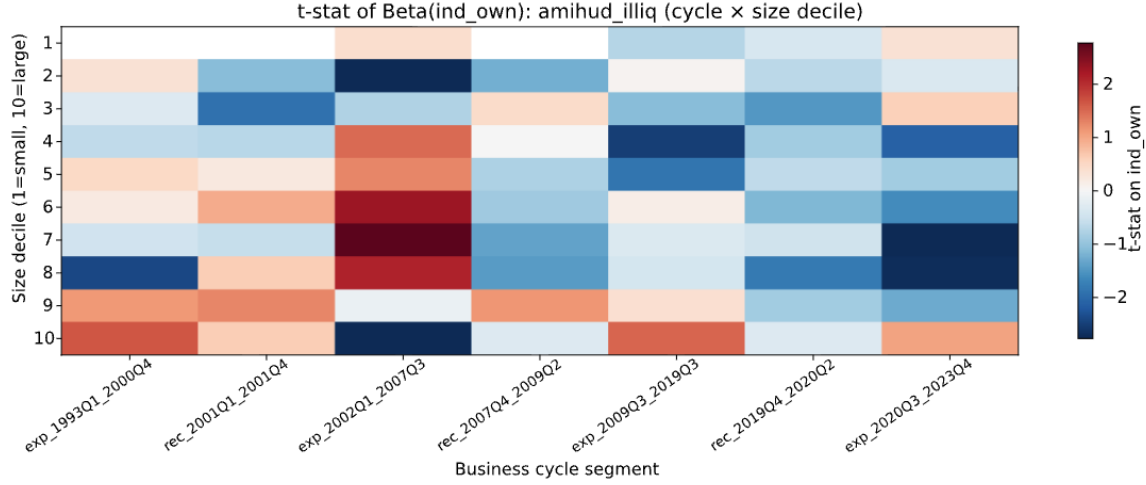


Figure 2.8: Size–Cycle Heatmap: Amihud Illiquidity

Notes: Heatmap of size-by-cycle coefficients for index ownership on Amihud illiquidity.

During recession windows, liquidity estimates are generally less precise because of shorter samples, and the signs vary by decile. The lack of uniformity across recession episodes indicates that liquidity effects are highly state dependent and may reflect a mixture of passive rebalancing and market-wide stress. This heterogeneity reinforces the earlier conclusion that liquidity is the most regime sensitive channel and that its relationship with ownership should not be summarized by a single average coefficient.

Volatility effects also vary across the joint size–cycle grid. The strongest cell appears in the 2001 recession for size decile 7, where ownership is associated with lower idiosyncratic volatility ($\beta = -1.8811$, $t = -4.46$). This pattern indicates that in early recession conditions, passive ownership coincided with a pronounced reduction in volatility for mid-sized firms. By contrast, many expansion-period cells show small or insignificant coefficients, suggesting that volatility responses are more concentrated in specific regime–size combinations. Figure 2.9 summarizes these results across all size–cycle cells.

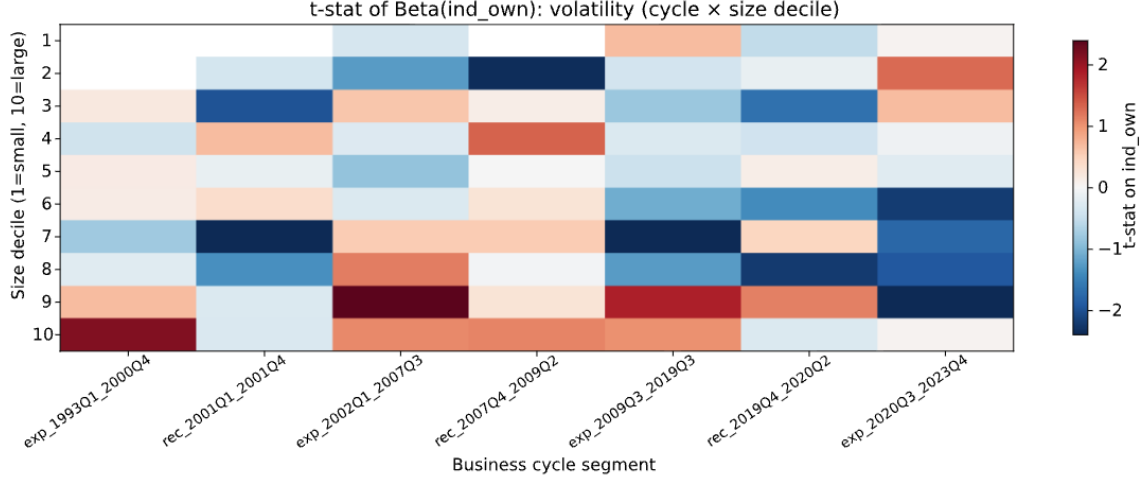


Figure 2.9: Size–Cycle Heatmap: Volatility

Notes: Heatmap of size-by-cycle coefficients for index ownership on idiosyncratic volatility.

The concentration of significant effects in select recession windows is consistent with the view that volatility responses depend on both firm size and macro stress. When aggregate shocks are large, ownership structure may affect how strongly firm-level volatility is dampened by passive flows, and this effect appears most visible among mid-sized firms. However, the lack of pervasive significance across the grid underscores that the volatility channel is episodic rather than universal. This point cautions against extrapolating any single recession estimate to all cycles or all firm sizes.

The price informativeness proxy displays the most robust and persistent negative association with index ownership across the size–cycle grid. The strongest cell occurs during the 2009Q3–2019Q3 expansion for size decile 4, where the coefficient is large and negative ($\beta = -18.0537$, $t = -9.04$). Across many expansion windows, mid-sized and large firms show consistently negative and significant coefficients, indicating that the informativeness channel remains a central trade-off in periods of normal growth. In recession windows, coefficients are generally less precise but do not overturn the

negative association observed in expansions. Figure 2.10 reports the full heatmap of informativeness estimates.

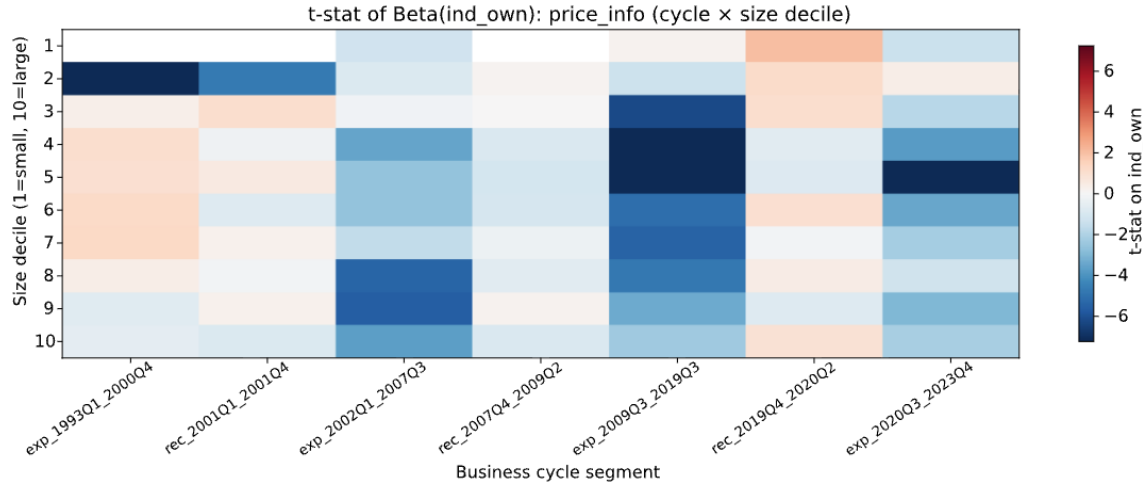


Figure 2.10: Size–Cycle Heatmap: Price Informativeness

Notes: Heatmap of size-by-cycle coefficients for index ownership on the price informativeness proxy.

The persistence of negative coefficients across size and cycle states suggests that the information-production channel is stable in direction, although its magnitude varies across environments. This finding aligns with the earlier business-cycle interaction results, where the ownership–informativeness slope did not materially differ between recessions and expansions. Nevertheless, the joint size-cycle heatmap shows that the largest negative estimates cluster during the 2009Q3–2019Q3 post-recession expansion, particularly among mid-sized firms. This pattern suggests that the ownership–informativeness association may be stronger during recovery periods, even though its direction remains broadly negative across the sample. Together, these results indicate that both macroeconomic conditions and firm size shape the strength of the relationship.

The joint size-cycle estimates can also be interpreted in terms of the change in each market-quality outcome associated with a one-percentage-point increase in index

ownership. Because the outcomes use different units, the coefficients should not be compared mechanically. For price informativeness, the coefficient of -18.0537 for mid-sized firms during the 2009Q3–2019Q3 expansion implies an estimated decline of approximately 0.18 units. The coefficient of -1.8811 for mid-sized firms during the 2001 recession implies a decline of approximately 0.019 units in idiosyncratic volatility, while the liquidity coefficient of 0.0348 implies an increase of approximately 0.00035 in Amihud illiquidity.

The results imply that the market-quality consequences of index ownership are uneven across channels. The negative association between index ownership and price informativeness is the broadest pattern, whereas potential benefits for volatility and liquidity are conditional: volatility reductions are concentrated among mid-sized firms during selected stressed periods, and liquidity effects appear only in particular size-cycle cells. Thus, the evidence does not support the view that greater index ownership uniformly improves or worsens market quality. Instead, it suggests that index ownership is consistently associated with an information-production trade-off, while its effects on trading conditions depend on where and when ownership changes occur. Because the analysis is descriptive, this pattern identifies an economically meaningful set of trade-offs rather than a causal optimum.

The joint design also offers a framework for connecting the cross-sectional and time-series evidence. If liquidity improvements are concentrated in certain size-cycle cells, then event-based or IV estimates should be interpreted relative to the macro regime and size composition of the affected firms. Similarly, persistent informativeness declines across expansions imply that ownership changes may carry information costs even in benign environments.

2.5 CONCLUSION

This chapter provides a structured heterogeneity analysis of how index ownership relates to market quality across firms and macroeconomic conditions. The size-sorted results show that the negative price-informativeness association is strongest among smaller and mid-sized firms, whereas volatility reductions are concentrated among mid-cap firms and liquidity effects are comparatively weak and uneven. The business-cycle analysis identifies liquidity as the most state-dependent channel, while volatility reductions weaken in recessions and the negative informativeness association remains present across regimes. The joint size-cycle analysis further shows that the negative informativeness association is especially pronounced during the 2009Q3–2019Q3 post-recession expansion, particularly among mid-sized firms, indicating that its direction is persistent but its magnitude varies across environments.

Taken together, the results point to an asymmetric market-quality trade-off. The negative association with price informativeness is broad and persistent, while lower volatility appears mainly in selected mid-cap stress environments and liquidity effects are localized to particular size-cycle conditions. The evidence therefore does not support a uniform market-quality effect or a single optimal ownership level. Instead, it identifies an information-production trade-off that is relatively general in direction, alongside trading-condition effects that depend on firm size and macroeconomic conditions.

However, there are 3 key limitations. The results remain descriptive and therefore cannot establish causal mechanisms. Second, some size-cycle cells are thin, especially in short recession windows, which reduces precision and increases sensitivity to sampling variation. The business-cycle indicator is binary and may mask variation within expansions or within downturns. Future work should combine the joint

heterogeneity framework with quasi-experimental ownership shocks and continuous financial-condition measures to assess whether these patterns persist under causal identification.

Despite these limitations, the heterogenous analysis is informative. The evidence implies that market-quality assessments should account for both firm composition and macro conditions. Information-production incentives appear most sensitive to passive ownership among smaller firms, while liquidity and volatility responses are more cyclical. This context dependence suggests that policy discussions should focus on marginal ownership changes in specific environments rather than on an overall average effect.

APPENDIX A

ADDITIONAL RESULTS FOR CHAPTER 1

A.1 RUSSELL INDEX CONSTRUCTION AND ASSIGNMENT

This appendix section documents the construction of the Russell market capitalization ranking used to assign firms to the Russell 1000 and Russell 2000 indices. The procedure follows the replication approach in Ben-David, Franzoni, and Moussawi (2018), which mirrors the annual reconstitution methodology used by FTSE Russell. The objective is to recover a consistent rank ordering when official rankings are not publicly available for the full sample period. The resulting ranks provide a transparent mapping from market capitalization to index assignment, which is central to the empirical design.

Russell indices are reconstituted annually at the end of June, and index membership is determined by market capitalization measured on the rank date, which typically occurs in late May. Firms are ordered by their market capitalization on the rank date, and the largest 1,000 firms are assigned to the Russell 1000, while the next 2,000 firms are assigned to the Russell 2000. This annual sorting produces a sharp change in index weights at the cutoff between ranks 1,000 and 1,001. As a result, firms near the cutoff are similar in size but face materially different index ownership due to benchmark tracking. The ranking universe is restricted to U.S. common stocks with ordinary share codes and primary exchange listings consistent with the standard

CRSP universe. Firms with non-common share codes, non-primary listings, or missing share and price data on the rank date are excluded to ensure that the ranking reflects the eligible investable set. When a firm has multiple share classes, the ranking is based on the primary class observed in CRSP so that market capitalization is not double-counted across classes. These screens align the replicated universe with the investable universe implied by the index methodology and reduce mechanical noise in the rank ordering.

Market capitalization is computed from CRSP daily data using the standard measure of price times shares outstanding. Specifically, for firm i in year t , market capitalization equals the absolute value of price times shares outstanding on the rank date. Because the official rank date is in late May, the last trading day of May is used to approximate the ranking input. This alignment ensures that the rank ordering reflects the information set used in index assignment while preserving a consistent data definition across years. The absolute value of price is used to accommodate CRSP's sign convention for bid-ask midpoint pricing, and shares outstanding are measured in thousands and scaled accordingly. For firms with missing data on the last trading day of May, the most recent prior trading day within May is used to avoid losing observations due to short-lived data gaps. Market capitalization is computed on an unadjusted basis to mirror index construction and to preserve the correct relative ranking across firms. These choices follow common practice in the Russell replication literature and minimize divergence from the official ranking input.

All eligible U.S. common stocks are sorted in descending order by market capitalization on the rank date. The ranking assigns rank 1 to the largest firm and proceeds sequentially through the eligible universe. Index assignment follows the published thresholds: ranks 1 through 1,000 are mapped to the Russell 1000 and ranks 1,001 through 3,000 are mapped to the Russell 2000. This procedure yields a deterministic

index classification that can be reproduced year by year from the underlying CRSP inputs.

Ties in market capitalization are rare at the upper tail, but when they occur the tie is resolved by the CRSP market equity ordering, which is equivalent to the raw ranking by market capitalization. The procedure uses the full ranked list to preserve relative ordering even when firms fall outside the top 3,000, ensuring that the cutoff firms are ranked within a consistent global ordering. Firms entering or exiting the universe between May and June are assigned based on the May rank-date information, consistent with the index methodology that fixes membership at reconstitution. This timing convention maintains a clean separation between ranking inputs and subsequent index membership outcomes.

Because official Russell rankings are not public for the full sample, this study replicates the ranking using CRSP market capitalization data. Following Ben-David, Franzoni, and Moussawi (2018), the replicated ranks are constructed from the rank-date market capitalization and then mapped to the Russell 1000 and Russell 2000 cutoffs. The replicated rankings closely match the index assignments reported in Russell membership files, and the approach is widely used in the literature on index inclusion. Accordingly, the replicated cutoff provides a credible and transparent basis for analyzing discontinuities in index ownership around the 1,000/1,001 threshold.

A.2 LOCAL IV ROBUSTNESS CHECKS

This appendix section evaluates the robustness of the IV results to alternative bandwidth choices around the Russell reconstitution cutoffs. The purpose is to test whether the local-window IV estimates confirm the global results or instead reveal weak-instrument behavior. Because identification relies on a discrete ownership jump at

the cutoff, narrowing the window should improve comparability but reduce first-stage variation. The analysis therefore focuses on how relevance, precision, and coefficient stability change across bandwidths. These diagnostics are reported for both the 1,000 and 2,000 cutoffs to assess sensitivity to the source of variation.

The robustness grid sweeps bandwidths of 50, 100, 150, 200, 300, and 500 ranks, in addition to the global sample, for each cutoff. For each window, the same 2SLS specification is re-estimated and diagnostics are recorded, including the first-stage F statistic and the index-ownership coefficient. The outputs include a numerical summary table and two plots that track first-stage strength and coefficient stability across bandwidths. All estimations use the same controls and fixed effects as the baseline IV design to ensure comparability. This grid-based design makes it possible to visualize how identification degrades as the window narrows.

First-stage strength deteriorates sharply in local windows. Across both cutoffs, F statistics fall well below conventional thresholds once the sample is restricted to narrow bandwidths, often below 1 and rarely above 5. By contrast, the global samples deliver strong first stages, with $F > 25$ at the 1,000 cutoff and $F > 130$ at the 2,000 cutoff. This pattern indicates weak identification in local variants and strong relevance only in global specifications. The difference in F statistics across cutoffs also reflects the larger ownership discontinuity at the 2,000 boundary.

Coefficient estimates are correspondingly unstable in local windows. The 2SLS coefficients on index ownership swing widely across narrow bandwidths and often exhibit large standard errors. In several windows the sign flips, which is consistent with weak instruments rather than true economic reversals. By comparison, global estimates are smoother and more interpretable, mirroring their stronger first-stage diagnostics. The instability is especially pronounced for volatility and price informativeness, where local samples are smallest.

Table A.1: IV Robustness Grid: Russell 1000

	50	100	150	200	300	500	global
Amihud illiquidity	0.0744 (0.0563)	0.0067 (0.0220)	0.0924 (0.1595)	0.0050 (0.0370)	0.0474 (0.5740)	0.4971 (2.2070)	0.2319 (0.1450)
	F=2.46 p=0.1169	F=2.16 p=0.1412	F=0.43 p=0.5111	F=2.14 p=0.1432	F=0.02 p=0.8861	F=0.05 p=0.8170	F=27.79 p=1.35e-07
	2.4288 (3.3057)	3.8671 (3.3886)	13.7755 (22.5574)	4.1938 (4.0521)	-10.8720 (73.8257)	20.9380 (98.3811)	-1.4850*** (0.4090)
Volatility	F=2.45 p=0.1172	F=2.19 p=0.1391	F=0.43 p=0.5132	F=2.15 p=0.1423	F=0.02 p=0.8774	F=0.05 p=0.8316	F=27.30 p=1.74e-07
	21.3384 (52.7615)	16.8883 (35.3193)	29.6162 (111.4591)	21.7115 (41.5191)	322.2773 (1071.8428)	109.6254 (543.8555)	-5.8808 (4.0159)
	F=2.42 p=0.1196	F=2.30 p=0.1291	F=0.39 p=0.5339	F=2.26 p=0.1324	F=0.10 p=0.7560	F=0.05 p=0.8225	F=28.49 p=9.44e-08
Price informativeness							
N	3009	6135	9081	12432	19556	36777	235544
Firms	539	791	950	1068	1276	1761	8627
Dates	99	99	99	99	99	99	99
Clusters	539	791	950	1068	1276	1761	8627
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Firm + time	Firm + time	Firm + time	Firm + time	Firm + time	Firm + time	Firm + time
SE type	Firm clustered	Firm clustered	Firm clustered	Firm clustered	Firm clustered	Firm clustered	Firm clustered

Notes: This table reports the quadratic IV robustness grid for the local-window specifications at the Russell 1000 cutoff. Each cell reports the 2SLS coefficient on *'ind_own'*, its standard error, and the first-stage F statistic / p-value. Bandwidth labels are in basis points; 'global' denotes the unrestricted sample. * p<0.10, ** p<0.05, *** p<0.01. Weak local-window cases should be interpreted cautiously, especially when the first-stage F statistic is far below the conventional threshold of 10.

Table A.2: IV Robustness Grid: Russell 2000

	50	100	150	200	300	500	global
Amihud illiquidity	-3.9920 (33.3665)	0.6955 (3.8003)	0.0153 (0.2220)	0.4610 (2.0254)	-0.2935 (0.5156)	-0.4358 (0.5759)	-0.2849*** (0.0776)
	F=0.01 p=0.9045	F=0.04 p=0.8434	F=1.42 p=0.2335	F=0.07 p=0.7967	F=0.79 p=0.3739	F=0.81 p=0.3673	F=131.16 p=<1e-16
	-0.5604 (22.8979)	1.1773 (13.9026)	-0.7824 (2.2575)	6.9003 (34.4527)	1.3420 (3.8259)	-3.7182 (5.1797)	-0.1767 (0.1263)
Volatility	F=0.01 p=0.9159	F=0.05 p=0.8259	F=1.47 p=0.2250	F=0.05 p=0.8304	F=0.70 p=0.4037	F=0.75 p=0.3859	F=128.98 p=<1e-16
	93.0678 (784.7418)	-108.9001 (520.3633)	-3.9988 (34.0028)	115.0171 (502.9717)	15.6064 (45.6625)	32.3535 (52.5557)	-12.0868*** (1.7568)
	F=0.02 p=0.8933	F=0.05 p=0.8297	F=1.31 p=0.2515	F=0.06 p=0.8045	F=0.90 p=0.3419	F=0.90 p=0.3440	F=131.12 p=<1e-16
Price informativeness							
N	4947	10458	15777	21039	30804	49683	235544
Firms	937	1448	1786	2015	2344	2877	8627
Dates	99	99	99	99	99	99	99
Clusters	937	1448	1786	2015	2344	2877	8627
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Firm + time	Firm + time	Firm + time	Firm + time	Firm + time	Firm + time	Firm + time
SE type	Firm clustered	Firm clustered	Firm clustered	Firm clustered	Firm clustered	Firm clustered	Firm clustered

Notes: This table reports the quadratic IV robustness grid for the local-window specifications at the Russell 2000 cutoff. Each cell reports the 2SLS coefficient on *'ind_own'*, its standard error, and the first-stage F statistic / p-value. Bandwidth labels are in basis points; 'global' denotes the unrestricted sample. * p<0.10, ** p<0.05, *** p<0.01. Weak local-window cases should be interpreted cautiously, especially when the first-stage F statistic is far below the conventional threshold of 10.

The robustness grid illustrates a standard trade-off between local identification credibility and statistical power. Local windows strengthen the exclusion restriction by comparing nearly identical firms but undermine relevance because the ownership jump becomes small. In this dataset, that relevance loss dominates, making local estimates too noisy for inference. Consequently, the main IV conclusions rely on the global cutoffs, with local results retained as qualitative checks. This approach is consistent with the broader literature on index reconstitution designs that faces similar power limitations.

A.3 QUADRATIC IV ROBUSTNESS CHECKS

This appendix section documents the nonlinear IV robustness checks using the Russell 1000 cutoff. The goal is to evaluate whether the quadratic IV results are stable when the source of exogenous variation is shifted from the 2000 cutoff to the 1000 cutoff. Because the ownership discontinuity at the 1000 threshold is smaller, these specifications are expected to be more fragile. The results are therefore reported as robustness evidence rather than as primary causal estimates.

The nonlinear IV design instruments both ind_own and ind_own^2 using the reconstitution indicator interacted with polynomial functions of the running variable. Controls include firm size and dollar volume as well as the running variable and its square, with firm and time fixed effects. This design mirrors the preferred cutoff-2000 specification but substitutes the 1000 cutoff as the source of identification. The same estimation protocol is applied to facilitate comparability across cutoffs.

The cutoff-1000 nonlinear IV diagnostics are unstable and not economically interpretable. First-stage statistics for ind_own and ind_own^2 become erratic, with signs of numerical instability in parts of the workflow. These problems indicate weak iden-

tification for the quadratic terms at the 1000 cutoff. As a result, coefficient estimates are sensitive to specification choices and should

The instability of the 1000-cutoff quadratic IV design is consistent with a smaller ownership discontinuity at the large-cap boundary. With limited variation in ownership, identifying both the level and square terms becomes especially difficult, amplifying weak-instrument concerns. Consequently, the nonlinear IV evidence relies on the 2000-cutoff specification, which delivers strong first-stage diagnostics. The 1000-cutoff results are retained here as a robustness appendix to demonstrate the sensitivity of nonlinear IV inference to instrument strength.

A.4 ADDITIONAL TABLES

Table A.3: Quadratic IV Comparison: Russell 1000

	b1	b2	Turning point	In support	F(own)	F(own ²)
Amihud illiquidity	0.4557	-2.7663*	0.0824	Yes	-5878.688	-2867.905
Volatility	-3.0590***	14.6507***	0.1044	Yes	-1994.040	552.483
Price informativeness	0.8048	-71.3398*	0.0056	No	-4238.726	-1945.796

Controls: `c_firm_size`, `c_dollar_vol`, `running`, and `running`². Firm and time fixed effects. Firm-clustered standard errors. Sample size is not reported in this comparison CSV.

Notes: This table reports the quadratic IV comparison for the global design at the Russell 1000 cutoff. Coefficient entries are reported with significance stars based on p-values. The turning-point indicator reports whether the estimated turning point lies inside the [q01, q99] support band. * p<0.10, ** p<0.05, *** p<0.01.

The first-stage columns report the joint F statistics for `ind_own` and `ind_own`². Sample size is not reported in this comparison CSV.

Table A.4: Quadratic IV Comparison: Russell 2000

	b1	b2	Turning point	In support	F(own)	F(own ²)
Amihud illiquidity	0.3313	-4.8236**	0.0343	Yes	638.786	28.413
Volatility	-2.5336***	15.6492***	0.0809	Yes	611.060	25.592
Price informativeness	-6.8010	-35.9083	-0.0947	No	664.395	34.308

Controls: *c_firm_size*, *c_dollar_vol*, *running*, and *running*². Firm and time fixed effects. Firm-clustered standard errors. Sample size is not reported in this comparison CSV.

Notes: This table reports the quadratic IV comparison for the global design at the Russell 2000 cutoff. Coefficient entries are reported with significance stars based on p-values. The turning-point indicator reports whether the estimated turning point lies inside the [q01, q99] support band. * p<0.10, ** p<0.05, *** p<0.01. The first-stage columns report the joint F statistics for *ind_own* and *ind_own*². Sample size is not reported in this comparison CSV.

Table A.5: Quadratic Nonlinearity Results

	Amihud illiquidity	Volatility	Price informativeness
Linear term (b_1)	0.0107 (0.0333)	-0.3250*** (0.0626)	-10.3568*** (1.1805)
Quadratic term (b_2)	0.1882** (0.0947)	0.9991*** (0.2076)	26.7002*** (4.8399)
Turning point	-0.0284 (0.1002)	0.1626 (0.0134)	0.1939 (0.0143)
In support	No	Yes	Yes
Observations	241199	240515	239758
Entities	8877	8858	8855
Time periods	100	100	100
Support min	0.0098	0.0098	0.0098
Support q01	0.0104	0.0104	0.0104
Support q05	0.0128	0.0128	0.0129
Support q50	0.0510	0.0511	0.0513
Support q95	0.1821	0.1822	0.1822
Support q99	0.2267	0.2268	0.2266
Support max	0.9526	0.9526	0.9526

Notes: This table reports the reduced-form quadratic fixed-effects results for the three market-quality outcomes. Coefficients are reported with standard errors underneath in parentheses, and significance stars are based on p-values. The turning-point row reports the implied optimum of the fitted quadratic and its standard error. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The support rows summarize the empirical distribution of ‘*ind_own*’ used to interpret whether the turning point lies inside the observed range.

Table A.6: Spline Panel Regression Results

	Amihud illiquidity	Volatility	Price informativeness
Spline basis 1	-0.0103*** (0.0018)	-0.0022* (0.0013)	-0.1057*** (0.0165)
Spline basis 2	-0.0123*** (0.0017)	-0.0142*** (0.0010)	-0.2848*** (0.0145)
Spline basis 3	-0.0089*** (0.0022)	-0.0266*** (0.0033)	-0.7690*** (0.0323)
Spline basis 4	0.0039 (0.0028)	-0.0129*** (0.0038)	-0.5774*** (0.0409)
Observations	241199	240515	239758
Entities	8877	8858	8855
Time periods	100	100	100
Within R^2	0.0898	0.2200	0.0132
Best ' <i>ind_own</i> '	0.0467	0.0733	0.0104
Best curve value	-0.0002	-0.0050	0.8715
Support q01	0.0104	0.0104	0.0104
Support q50	0.0510	0.0511	0.0513
Support q99	0.2267	0.2268	0.2266

Notes: This table reports spline panel regressions with firm and time fixed effects. Standard errors are reported underneath coefficients in parentheses. The spline basis is a natural cubic spline in '*ind_own*' with four basis terms. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The sample support rows report the empirical ownership distribution used for interpreting the fitted spline.

Table A.7: IV Regression Results for Spline IV at Global Russell 2000 Cutoff

	Amihud illiquidity	Volatility	Price informativeness
Spline basis 1	0.0095 (0.0191)	-0.0086 (0.0269)	-0.3619 (0.3264)
Spline basis 2	0.0095** (0.0038)	-0.0068 (0.0072)	-0.3663*** (0.0964)
Spline basis 3	-0.0040 (0.0440)	0.0343 (0.0781)	-1.0792 (1.0048)
Spline basis 4	-0.0093 (0.0487)	0.0093 (0.0581)	-0.4132 (0.6949)
Observations	235544	234897	234281
Firms	8627	8605	8614
Dates	99	99	99
Cutoff	2000	2000	2000
Poly degree	4	4	4
Min first-stage F	15.7369	15.9077	15.6617
Best ind_own	22.7%	3.3%	14.7%
Controls	Yes	Yes	Yes
Fixed effects	firm + date	firm + date	firm + date
SE type	firm clustered	firm clustered	firm clustered

Notes: This table reports the second-stage IV spline coefficients for the global Russell 2000 design. Standard errors are reported underneath coefficients in parentheses. The dependent-variable panels correspond to Amihud illiquidity, volatility, and price informativeness. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The spline basis is a natural cubic spline in **ind_own** with firm and date fixed effects and firm-clustered standard errors.

APPENDIX B

ADDITIONAL RESULTS FOR CHAPTER 2

B.1 ADDITIONAL TABLES

Table B.1: Size-Sorted Regression Results by Market Capitalization Decile

(a) Panel A: Amihud illiquidity				(b) Panel B: Volatility			
Decile	Ind. own.	Firm size	Dol. vol.	Decile	Ind. own.	Firm size	Dol. vol.
1	-0.0678 (0.1017)	-0.1330*** (0.0141)	-0.0566*** (0.0027)	1	-0.2098* (0.1089)	-0.1129*** (0.0072)	0.0835*** (0.0037)
2	-0.0264 (0.0193)	-0.0175*** (0.0016)	-0.0173*** (0.0008)	2	-0.0730 (0.0710)	-0.1071*** (0.0072)	0.0769*** (0.0041)
3	0.0191** (0.0084)	-0.0089*** (0.0007)	-0.0079*** (0.0004)	3	-0.0157 (0.0606)	-0.1255*** (0.0077)	0.0805*** (0.0034)
4	0.0007 (0.0027)	-0.0031*** (0.0004)	-0.0042*** (0.0003)	4	-0.1395** (0.0668)	-0.1100*** (0.0058)	0.0790*** (0.0028)
5	-0.0006 (0.0017)	-0.0019*** (0.0002)	-0.0023*** (0.0001)	5	-0.1255** (0.0637)	-0.1234*** (0.0063)	0.0828*** (0.0035)
6	-0.0027 (0.0025)	-0.0006** (0.0003)	-0.0017*** (0.0003)	6	-0.1060** (0.0469)	-0.1037*** (0.0056)	0.0768*** (0.0028)
7	-0.0013 (0.0010)	-0.0002 (0.0002)	-0.0009*** (0.0001)	7	-0.0430 (0.0394)	-0.0960*** (0.0061)	0.0728*** (0.0042)
8	0.0003 (0.0007)	-0.0001 (0.0001)	-0.0005*** (0.0002)	8	0.0407 (0.0544)	-0.0928*** (0.0049)	0.0748*** (0.0034)
9	0.0004 (0.0003)	-0.0000 (0.0000)	-0.0003*** (0.0001)	9	-0.0449 (0.0520)	-0.0811*** (0.0044)	0.0769*** (0.0035)
10	0.0002 (0.0001)	0.0000 (0.0000)	-0.0001*** (0.0000)	10	0.0414 (0.0374)	-0.0620*** (0.0044)	0.0699*** (0.0048)
Observations	23929			Observations	23929		
within R^2	0.2542			within R^2	0.2674		

Table B.1 (continued)**(a) Panel C: Price informativeness**

Decile	Ind. own.	Firm size	Dol. vol.
1	-13.5969*** (1.2993)	-0.5095*** (0.0474)	-0.0180 (0.0149)
2	-8.2094*** (0.6583)	-0.6022*** (0.0537)	0.2268*** (0.0177)
3	-3.8004*** (0.5556)	-0.6176*** (0.0713)	0.2643*** (0.0219)
4	-3.5096*** (0.4750)	-0.5963*** (0.0642)	0.3350*** (0.0240)
5	-3.7967*** (0.5085)	-0.5927*** (0.0607)	0.4377*** (0.0296)
6	-2.4811*** (0.4838)	-0.4569*** (0.0637)	0.4591*** (0.0283)
7	-2.4157*** (0.4029)	-0.2969*** (0.0580)	0.4501*** (0.0288)
8	-2.8178*** (0.5606)	-0.3852*** (0.0548)	0.4437*** (0.0348)
9	-2.2610*** (0.6090)	-0.3975*** (0.0538)	0.4666*** (0.0451)
10	-0.4039 (0.8068)	-0.4402*** (0.0620)	0.4068*** (0.0546)
Observations	23929		
within R^2	0.0714		

Notes: This table reports the size-sorted regression results in three panels. Standard errors are reported underneath coefficients in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Sample construction: deciles are based on market capitalization. Regressions are estimated separately within each decile for each outcome variable.

Table B.2: Heterogeneity by Recession: Panel FE with Interaction

	Amihud illiquidity	Volatility	Price informativeness
Industry ownership	0.0800*** (0.0237)	-0.0681*** (0.0231)	-3.2530*** (0.2401)
Industry ownership $\times$ recession	-0.2142** (0.0867)	0.1390 (0.1036)	-0.5590 (0.8015)
Firm size	0.0082*** (0.0016)	-0.0983*** (0.0031)	-0.3411*** (0.0267)
Dollar volume	-0.0221*** (0.0017)	0.0747*** (0.0024)	0.1456*** (0.0147)
Observations	241199	240515	239758
Entities	8877	8858	8855
Time periods	100	100	100
Firm FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Two-way clustered SE	Yes	Yes	Yes

Notes: This table reports panel fixed-effects regressions with an interaction between industry ownership and recession periods. Standard errors are reported underneath coefficients in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The interaction coefficient is the incremental slope in recession quarters relative to non-recession quarters. Standard errors are two-way clustered by firm and date.

Table B.3: Business-Cycle by Size-Decile Regression Results: Amihud illiquidity

Size decile	Exp. 1993-2000	Rec. 2001	Exp. 2002-2007	Rec. 2007-2009	Exp. 2009-2019	Rec. 2019-2020	Exp. 2020-2023
1	—	—	4.1130 (10.2765)	—	-6.7137 (9.1838)	-5.4107 (14.0474)	0.4476 (1.3706)
2	0.6909 (2.0553)	-60.5637 (55.0788)	-6.0882*** (1.6093)	-35.2844 (28.3178)	0.0602 (0.9387)	-1.1240 (1.6690)	-0.1624 (0.4595)
3	-1.8919 (5.8612)	-18.7479* (9.7675)	-0.6591 (0.8669)	1.8330 (4.0609)	-0.3110 (0.2847)	-1.7923 (1.2064)	0.0948 (0.1564)
4	-0.5965 (0.9196)	-2.2226 (3.1872)	0.3133 (0.2099)	-0.0062 (0.4088)	-0.1050** (0.0417)	-0.2233 (0.2567)	-0.2077** (0.0980)
5	0.8900 (1.8452)	0.2236 (0.9405)	0.0904 (0.0714)	-0.1004 (0.1287)	-0.0560* (0.0300)	-0.0944 (0.1477)	-0.0315 (0.0361)
6	0.0851 (0.4276)	0.4979 (0.5302)	0.0947** (0.0414)	-0.0479 (0.0531)	0.0016 (0.0111)	-0.0641 (0.0558)	-0.0225 (0.0140)
7	-0.0483 (0.1061)	-0.0231 (0.0388)	0.0348*** (0.0097)	-0.0230 (0.0168)	-0.0008 (0.0022)	-0.0044 (0.0091)	-0.0115*** (0.0041)
8	-0.0857** (0.0356)	0.0159 (0.0263)	0.0095** (0.0045)	-0.0076 (0.0053)	-0.0015 (0.0035)	-0.0122* (0.0067)	-0.0077*** (0.0028)
9	0.0318 (0.0287)	0.0381 (0.0303)	-0.0003 (0.0023)	0.0031 (0.0028)	0.0002 (0.0006)	-0.0028 (0.0032)	-0.0031 (0.0024)
10	0.0079* (0.0047)	0.0013 (0.0021)	-0.0015*** (0.0005)	-0.0002 (0.0008)	0.0002 (0.0001)	-0.0004 (0.0012)	0.0004 (0.0003)

Notes: Each cell reports the coefficient on industry ownership from a two-way fixed-effects regression estimated within a business-cycle segment and size decile. Standard errors are reported underneath coefficients in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

The cycle labels are abbreviated as expansion (Exp.) and recession (Rec.). Cell-level sample sizes, entity counts, and time-period counts vary across regressions and are stored in the source CSV. A dash indicates that the source regression cell is not available.

Table B.4: Business-Cycle by Size-Decile Regression Results: Volatility

Size decile	Exp. 1993-2000	Rec. 2001	Exp. 2002-2007	Rec. 2007-2009	Exp. 2009-2019	Rec. 2019-2020	Exp. 2020-2023
1	—	—	-1.6504 (4.7741)	—	1.3217 (1.9270)	-1.3644 (2.5443)	0.0181 (0.4618)
2	—	-4.4391 (12.4485)	-1.1188 (0.8961)	-13.3418** (5.7042)	-0.1360 (0.3636)	-0.2176 (1.5972)	0.5831 (0.4564)
3	1.3540 (6.5851)	-9.5938* (4.8497)	0.3931 (0.6450)	0.4814 (4.2206)	-0.1290 (0.1583)	-1.6612* (0.9987)	0.2015 (0.2941)
4	-0.4578 (1.1001)	3.1674 (4.6317)	-0.0971 (0.3556)	1.8182 (1.3683)	-0.0312 (0.1096)	-0.3983 (0.9718)	-0.0193 (0.2513)
5	0.6315 (3.8147)	-0.4871 (3.2858)	-0.1974 (0.2272)	-0.0011 (0.6172)	-0.0528 (0.1171)	0.1296 (1.0930)	-0.0433 (0.1787)
6	0.3384 (2.5666)	1.4813 (4.1616)	-0.0439 (0.1501)	0.1666 (0.6170)	-0.1032 (0.0951)	-1.7183 (1.2327)	-0.3089** (0.1408)
7	-0.9261 (1.2022)	-1.8811*** (0.4220)	0.0802 (0.1470)	0.2268 (0.4083)	-0.1376** (0.0571)	0.3013 (0.6618)	-0.5120* (0.2910)
8	-0.4373 (1.9467)	-1.2190 (0.8966)	0.1818 (0.1580)	-0.0100 (0.3470)	-0.0643 (0.0520)	-0.7590** (0.3415)	-0.2931* (0.1532)
9	1.6041 (2.3656)	-0.9216 (3.1151)	0.4801** (0.1905)	0.1322 (0.4928)	0.0685* (0.0371)	0.4798 (0.4240)	-0.2146*** (0.0737)
10	3.2393** (1.5136)	-0.5515 (1.7810)	0.2146 (0.1995)	0.9055 (0.8097)	0.0312 (0.0304)	-0.0823 (0.2779)	0.0026 (0.0597)

Notes: Each cell reports the coefficient on industry ownership from a two-way fixed-effects regression estimated within a business-cycle segment and size decile. Standard errors are reported underneath coefficients in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

The cycle labels are abbreviated as expansion (Exp.) and recession (Rec.). Cell-level sample sizes, entity counts, and time-period counts vary across regressions and are stored in the source CSV. A dash indicates that the source regression cell is not available.

Table B.5: Business-Cycle by Size-Decile Regression Results: Price informativeness

Size decile	Exp. 1993-2000	Rec. 2001	Exp. 2002-2007	Rec. 2007-2009	Exp. 2009-2019	Rec. 2019-2020	Exp. 2020-2023
1	—	—	-61.4089 (51.0435)	—	6.0464 (27.7230)	50.3437* (25.3792)	-11.0668 (7.5884)
2	-19.0526*** (1.7280)	-276.8213*** (58.8438)	-7.4068 (8.3050)	6.0342 (45.9908)	-12.9812 (9.6485)	24.0119 (20.8303)	1.8399 (5.3546)
3	9.4201 (30.5327)	70.0981 (68.2479)	-3.6210 (18.0952)	1.3959 (34.1508)	-18.5138*** (2.9544)	11.4248 (11.0397)	-8.3910* (4.5077)
4	13.2106 (12.5583)	-22.9894 (84.6787)	-27.4725*** (7.7381)	-22.0223 (23.5537)	-18.0537*** (1.9969)	-2.8037 (4.4130)	-11.7401*** (3.1138)
5	33.4625 (34.2266)	17.8546 (32.5824)	-12.7003*** (4.8500)	-8.7552 (8.1478)	-8.9452*** (1.1354)	-5.1195 (6.4164)	-11.0855*** (1.4486)
6	37.3708 (30.5184)	-25.3137 (33.4932)	-5.5851*** (2.1343)	-4.0640 (3.9576)	-4.1061*** (0.8037)	6.0145 (5.9116)	-5.3036*** (1.5333)
7	37.2193 (28.9148)	2.8618 (11.2145)	-3.6360 (2.2811)	-1.0371 (3.3058)	-4.0113*** (0.7319)	-0.7917 (5.9930)	-3.3728** (1.5294)
8	9.4079 (26.5372)	-1.2006 (10.5442)	-14.5853*** (2.7091)	-2.0700 (3.2440)	-3.0247*** (0.6342)	1.1919 (2.9233)	-1.2936 (0.9958)
9	-12.1230 (17.4200)	12.0239 (48.6792)	-16.3989*** (2.9173)	0.5630 (3.0757)	-2.0913*** (0.6218)	-2.0598 (2.6029)	-2.1893*** (0.7196)
10	-12.9384 (22.8037)	-34.1166 (39.1784)	-14.0360*** (3.7638)	-5.0692 (5.5135)	-1.7743** (0.7481)	2.8694 (3.3778)	-2.1335** (0.9855)

Notes: Each cell reports the coefficient on industry ownership from a two-way fixed-effects regression estimated within a business-cycle segment and size decile. Standard errors are reported underneath coefficients in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

The cycle labels are abbreviated as expansion (Exp.) and recession (Rec.). Cell-level sample sizes, entity counts, and time-period counts vary across regressions and are stored in the source CSV. A dash indicates that the source regression cell is not available.

Table B.6: Best Business-Cycle by Size-Decile Results

Panel A: Global top 15 significant cells						Panel B: Top 5 significant cells by dependent variable							
Rank	Dep. var.	Cycle	Decile	β (SE)	p	Score	Rank	Dep. var.	Cycle	Decile	β (SE)	p	Score
<i>Amihud illiquidity</i>													
1	Price informativeness	Exp. 2009-2019	4	-18.0537*** (1.9969)	0.0000	80.00	1	Exp.	2002-2007	7	0.0348*** (0.0097)	0.0003	32.37
2	Price informativeness	Exp. 2009-2019	5	-8.9452*** (1.1354)	0.0000	72.07	2	Exp.	2002-2007	10	-0.0015*** (0.0005)	0.0045	25.39
3	Price informativeness	Exp. 2020-2023	5	-11.0855*** (1.4486)	0.0000	61.22	3	Exp.	2020-2023	8	-0.0077*** (0.0028)	0.0067	24.10
4	Price informativeness	Exp. 1993-2000	2	-19.0526*** (1.7280)	0.0000	55.54	4	Exp.	2020-2023	7	-0.0115*** (0.0041)	0.0052	23.94
5	Price informativeness	Exp. 2009-2019	7	-4.0113*** (0.7319)	0.0000	52.29	5	Exp.	2002-2007	2	-6.0882*** (1.6093)	0.0002	23.93
6	Price informativeness	Exp. 2009-2019	3	-18.5138*** (2.9544)	0.0000	50.48	<i>Volatility</i>						
7	Price informativeness	Exp. 2002-2007	9	-16.3989*** (2.9173)	0.0000	49.91	6	Rec.	2001	7	-1.8811*** (0.4220)	0.0000	30.33
8	Price informativeness	Exp. 2002-2007	8	-14.5853*** (2.7091)	0.0000	48.49	7	Exp.	2020-2023	9	-0.2146*** (0.0737)	0.0036	26.84
9	Price informativeness	Exp. 2009-2019	6	-4.1061*** (0.8037)	0.0000	47.63	8	Exp.	2009-2019	7	-0.1376*** (0.0571)	0.0159	23.00
10	Price informativeness	Exp. 2009-2019	8	-3.0247*** (0.6342)	0.0000	46.60	9	Exp.	2002-2007	9	0.4801** (0.1905)	0.0118	22.38
11	Price informativeness	Exp. 2009-2019	9	-2.0913*** (0.6218)	0.0008	33.53	10	Exp.	2020-2023	6	-0.3089*** (0.1408)	0.0283	18.23
<i>Price informativeness</i>													
12	Price informativeness	Exp. 2002-2007	10	-14.0360*** (3.7638)	0.0002	33.33	11	Exp.	2009-2019	4	-18.0537*** (1.9969)	0.0000	80.00
13	Amihud illiquidity	Exp. 2002-2007	7	0.0348*** (0.0097)	0.0003	32.37	12	Exp.	2009-2019	5	-8.9452*** (1.1354)	0.0000	72.07
14	Volatility	Rec. 2001	7	-1.8811*** (0.4220)	0.0000	30.33	13	Exp.	2020-2023	5	-11.0855*** (1.4486)	0.0000	61.22
15	Price informativeness	Exp. 2020-2023	4	-11.7401*** (3.1138)	0.0002	29.42	14	Exp.	1993-2000	2	-19.0526*** (1.7280)	0.0000	55.54
							15	Exp.	2009-2019	7	-4.0113*** (0.7319)	0.0000	52.29

Notes: The table lists the highest-scoring significant coefficient estimates on industry ownership. Score is defined in the source workflow as $|t| \times \log(1 + N)$. Standard errors are reported underneath coefficients in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The global panel ranks all significant cells, while the dependent-variable panel keeps the top five cells within each outcome group.

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